

Market Concentration as a Signal for Tail Risk

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Abstract

This paper tests whether rising concentration in high-Domar-weight sectors predicts the repricing of tail risk in equity options markets. Using a monthly panel of eleven Select Sector SPDR ETFs (2005–2026), we estimate two-way fixed-effects predictive regressions of the one-month-ahead 25-delta risk reversal on a novel Input-Output-weighted Supplier HHI measure. Supplier HHI is the dominant predictor of downside risk ($\hat{\beta} = -0.076$, $t = -5.22$). Our sector-heterogeneous specifications recover a substitutability gradient that matches the non-linear amplification mechanism of Baqaee and Farhi (2019), where physical-capital-intensive sectors load positively on concentration, while digital sectors load negatively. These findings demonstrate that economic complementarities, not concentration alone, drive the systemic amplification of tail risk.

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1 Introduction

A handful of factories in Taiwan produce nearly all of the world’s most advanced computer chips, and a single Dutch company, ASML, is the only firm in the world capable of building the extreme-ultraviolet lithography machines required to etch the most advanced features onto those chips. If either node failed for a sustained period, no replacement is plausible on a timeline of months, let alone weeks. Smartphones, automobiles, data centers, and the artificial intelligence stack that depends on them would confront the same shortage at once. The dependency is not hypothetical. It is a feature of the production network the global economy has knowingly assembled over the past two decades, and it is the kind of dependency this paper argues should be visible in financial market prices.

Recent history supports the claim that such dependencies generate latent tail risk. When the Tōhoku earthquake struck Japan in March 2011, damage to a single Renesas semiconductor plant cascaded into shutdowns across the global automotive industry. Toyota and Honda halted North American production lines for inputs they could not source elsewhere on short notice. When the Ever Given grounded itself across the Suez Canal in March 2021, six days of blockage rippled through container shipping schedules for months. Each event illustrates a structural feature of modern production. When supply networks consolidate around a small number of essential nodes, the network is highly efficient most of the time and catastrophically fragile when a hub fails.

The same logic of concentration now characterizes the U.S. stock market itself. The seven largest firms account for roughly a third of the total value of the S&P 500, leaving the index as top-heavy as it has been at any point in the past half-century. This matters far beyond Wall Street because of the wealth effect, the well-documented tendency of households to spend more when their stock and retirement holdings rise and to pull back when those holdings fall. As a larger share of household savings has migrated into equities over the past generation, the link between stock prices and consumer spending has tightened. A

sharp reversal in those few dominant firms would translate quickly into a broad pullback in everyday spending across the economy. Whether the current configuration represents peak efficiency or a systemic vulnerability waiting to materialize is the question that motivates this paper, and the answer turns on whether financial markets correctly price the kind of crash risk that concentrated production networks create.

When policymakers worry about dominant firms, their concerns usually run to antitrust enforcement and the inflation that follows when a small number of firms can raise prices without losing customers. Most macroeconomic research on concentration follows the same template, focusing on how concentration affects aggregate output and the markups firms charge over their costs. The structural-fragility dimension has received less attention despite its growing macroeconomic relevance. The relevant question is whether concentrated production networks make the economy more vulnerable to rare but severe downside events that show up in real output and asset prices. Equity option markets are sophisticated machines for processing information about future volatility, and decades of empirical work confirm that they price symmetric risk well. Whether they also price the asymmetric, left-tail risk that arises specifically from concentrated production networks is much less settled. Asymmetric tail risk has a distinctive financial signature. When investors are more worried about a sharp downward move in a stock than they are excited about a sharp upward move, the cost of buying protection against a crash rises relative to the cost of buying upside exposure, and this gap is visible in option prices. Whether the gap shifts predictably with changes in real-economy concentration is the empirical question this paper takes up.

Investigating that question requires accessible measures of the relevant constructs. The Herfindahl-Hirschman Index, or HHI, quantifies how concentrated a sector is by summing the squared market-capitalization weights of the firms in the sector. An HHI close to one indicates that a sector's market value sits in a single firm; an HHI close to zero indicates that the value is spread evenly across many firms of similar size. The 25-delta risk reversal, the principal financial outcome studied here, is the difference between the implied volatility of an

out-of-the-money put option and the implied volatility of an out-of-the-money call option at matched strikes. In plain language, a more negative risk reversal means investors are paying a larger premium for crash insurance than for upside speculation, signaling heightened concern about a downside surprise. The 25-delta naming convention selects options whose strikes lie a standardized distance from the current price, ensuring comparability across sectors and time. We use the one-month-ahead risk reversal as the dependent variable throughout, so any documented relationship is genuinely predictive rather than coincident.

Macroeconomics now offers a well-developed theoretical framework connecting concentration to asymmetric tail risk. Gabaix (2011) showed that idiosyncratic shocks to large firms do not diversify away in economies with heavy-tailed firm-size distributions, so granular movements in heavyweight firms drive aggregate volatility. Acemoglu et al. (2012) formalized how the topology of the input-output network governs the propagation of these shocks, distinguishing balanced networks that absorb disturbances from star networks that channel them. Baqaee and Farhi (2019) sharpened the link to tail risk by showing that production complementarities (situations in which inputs cannot substitute for one another in the short run) break the linearity that would otherwise make output respond symmetrically to symmetric shocks. Under complementarity, negative shocks to a critical node are amplified through the chain while positive shocks are dampened by other rigidities, generating endogenous negative skewness in real output even from symmetric primitive disturbances. Appendix A contains a self-contained derivation of this curvature mechanism. We specialize the standard multi-sector model to a simplified single-tier setting in which the central second-derivative term reduces to $\lambda_i(1 - \lambda_i)(\sigma - 1)$, where λ_i is sector i 's share of total economic activity and σ measures how easily one input can substitute for another when supply is disrupted. Recent work by Acemoglu and Tahbaz-Salehi (2025) ports a phase-transition logic from financial networks to production networks, establishing that the relationship between concentration and aggregate fragility is not monotone. At low to moderate concentration, hub formation reduces vulnerability to ordinary shocks because central nodes diversify across many users.

Beyond a threshold, the same hubs become channels of contagion. The pricing implications of this non-monotonic structure for the equity option surface have not previously been tested.

We test whether rising concentration in critical, high-Domar-weight sectors leads the repricing of tail risk. The Domar weight of a sector is its nominal sales as a share of GDP, and concentration is high-Domar-weight when it occurs in sectors whose macroeconomic footprint is itself large enough that a hub failure carries aggregate consequences. The empirical exercise builds a monthly panel of the eleven Select Sector SPDR ETFs from January 2005 through February 2026, combining Bloomberg constituent-level data, OptionMetrics implied volatility surfaces, and BEA Input-Output tables. We then estimate a series of two-way fixed-effects predictive regressions of the forward 25-delta risk reversal on contemporaneous concentration measures.

The paper makes three contributions to the literatures on production networks and asset pricing. First, we propose Supplier HHI as a new, high-frequency, Input-Output-weighted measure of inherited upstream concentration. For each sector, Supplier HHI takes the concentration of every upstream sector and weights it by the share of inputs the downstream sector buys from that upstream sector. The construction differs from sector HHI, which measures internal consolidation, and from the firm-level concentration measures used in prior work. It captures the precise vulnerability that the TSMC and ASML examples illustrate, a downstream industry inheriting the fragility of a concentrated upstream hub on which it depends. Second, we bridge two literatures that have largely run in parallel. The production-networks tradition rooted in Acemoglu et al. (2012), Baqaee and Farhi (2019), and Acemoglu and Tahbaz-Salehi (2025) has documented the real-economy consequences of network concentration, while the asset-pricing-of-tail-risk tradition rooted in Bakshi et al. (2003) and Kozhan et al. (2013) has documented how tail risk is priced in option markets. Herskovic (2018) builds the natural bridge by showing that shocks to the input-output network represent a non-diversifiable, priced systematic risk factor. The present paper extends that bridge from the average expected-return moment to the asymmetric tail-pricing moment of the eq-

uity option surface. Third, our sector-heterogeneous specifications give empirical content to the prediction of Baqaee and Farhi (2019) that complementarities, not concentration alone, generate non-linear amplification of downside risk.

The findings recover a sharp economic gradient that the homogeneous specifications conceal. When the loading on internal sector concentration is allowed to vary across sectors, seven physical-capital-intensive sectors (Real Estate, Health Care, Utilities, Materials, Industrials, Energy, and Consumer Staples) load positively on HHI in predicting forward tail-risk pricing, while two digital sectors (Technology and Communication) load negatively. This is what the higher effective elasticity of substitution governing their input bundles predicts. Economic magnitudes range from roughly ten basis points of risk-reversal change per one-standard-deviation increase in HHI in Real Estate and Energy, down to negative or near-zero values in Technology and Communication. Figure 3 in Section 5 displays this substitutability gradient as the visual centerpiece of the empirical results. The pattern matches the theoretical prediction that complementarities, which are most binding in physical production networks with long lead times and specialized capital goods, drive the non-linear amplification mechanism that translates concentration into asymmetric tail risk. In sectors where digital inputs can be rerouted and platform infrastructure provides built-in redundancy, the same mechanism is muted, and the option market correctly registers the difference. The Input-Output-weighted Supplier HHI measure carries an independent and statistically robust negative loading in the homogeneous baseline, with a coefficient of -0.076 that is significant at the one-percent level. Read through the non-monotonic fragility framework, this loading reflects the variance-compression branch of the relationship in the cross-section average. The tail-amplification branch becomes visible only once sector heterogeneity is permitted.

These results carry direct implications for how risk managers and macroprudential supervisors should think about sectoral concentration. A given level of HHI in a Materials or Energy sector implies more latent left-tail exposure than the same HHI in a Technology or Communication sector, and the option market embeds this distinction in its forward

pricing. Concentration in physical-capital-intensive sectors and concentration in digital sectors are economically distinct objects and should not be aggregated into a single market-wide concentration statistic. Monitoring upstream Supplier HHI provides a complementary, slow-moving leading indicator that aggregates fragility information across the inter-sector dependency structure rather than within any single industry, capturing precisely the kind of vulnerability that TSMC and ASML exemplify in the contemporary semiconductor supply chain.

The remainder of the paper is organized as follows. Section 2 reviews the related literatures on production networks and the asset pricing of tail risk. Section 3 develops the theoretical framework, deriving the diagonal-curvature mechanism that links concentration to asymmetric tail risk and the non-monotonic phase-transition relation that governs the sign of the predictive loading. Section 4 describes the data, the construction of Supplier HHI, and the staged empirical strategy. Section 5 reports the in-sample, sector-heterogeneous, and out-of-sample results, develops the substitutability-gradient interpretation, and discusses the limitations and robustness of the findings. Section 6 concludes.

2 Literature Review

This study bridges the gap between “New Production Network” macroeconomics and the empirical asset pricing literature on tail risk. Existing research has thoroughly documented the impact of market concentration on aggregate output (GDP) and markups, but a critical gap remains in understanding how concentration signals future catastrophic downside risk in financial markets.

2.1 The Macroeconomic Origins of Network Fragility

To understand how financial markets price systemic tail risk, we must first examine the physical mechanisms that generate it inside the real economy. Carvalho and Tahbaz-Salehi (2019)

provide the standard survey of this rapidly developing literature, organizing the channels through which microeconomic disturbances propagate through input-output linkages into aggregate fluctuations. We use this orienting reference to situate the present paper within a mainstream production-networks tradition.

The theoretical foundation of this paper rests on the Granular Hypothesis of Gabaix (2011). Breaking with standard macroeconomic models that assume idiosyncratic shocks can be diversified away, Gabaix demonstrated that in an economy dominated by heavy-tailed firm-size distributions (Zipf’s Law), the idiosyncratic movements of the largest granular firms drive a significant share of aggregate volatility. This grounds our use of heavyweight-firm volatility (Hub Shock) as a non-diversifiable disturbance to which the options market should respond.

Size alone, however, does not explain fragility. Acemoglu et al. (2012) formalized the transmission mechanism through network topology, distinguishing between balanced networks (stable) and star networks (fragile). In a star network, central hub sectors (the semiconductor industry today, for example) act as universal inputs, so shocks to these nodes do not dissipate but cascade through the production chain and prevent diversification. We use this paper because it supplies the network-topology vocabulary, hubs and balanced networks, on which the rest of the literature review and the theory section depend.

The standard benchmark for evaluating the aggregate importance of individual sectors is Hulten (1978). The theorem states that, under competitive equilibrium and constant returns to scale, the elasticity of aggregate output with respect to a sector’s productivity shock equals that sector’s Domar weight (its nominal sales as a share of GDP). This first-order result implies that positive and negative shocks of equal magnitude have symmetric effects on output, which means it provides no mechanism for concentration to generate directional tail risk. Hulten’s Theorem matters here as the symmetric baseline that any directional-risk story must explicitly break. Recent work has pivoted this literature from pure topology toward structural “economic bottlenecks” by showing how complementarities break this symmetric

baseline. Baqaee and Farhi (2019) demonstrate that modern production networks are defined by strong complementarities (low substitutability), which invalidate the linearity of Hulten’s Theorem. In such networks, negative shocks are amplified as the bottleneck constrains the entire production chain, while positive shocks are attenuated by other network rigidities. The result is endogenous negative skewness in real output even when the underlying disturbances are symmetric. We rely on this result because it is the formal source of the diagonal-curvature derivation in Section 3.2 and the ultimate justification for predicting directional skewness from concentration.

Barrot and Sauvagnat (2016) empirically validate this physical asymmetry. They use natural disasters to show that when inputs are highly specific (low elasticity of substitution), idiosyncratic shocks like the sudden loss of an un-substitutable supplier propagate aggressively downstream and cause non-linear output drops in customer firms. This confirms empirically that the substitutability parameter σ , on which the theory section’s predictions hinge, is functioning in the real data. Elliott et al. (2022) extend the picture by showing that firms’ endogenous optimization of sourcing relationships can drive the economy into a fragile regime in which small shocks generate discontinuously large output drops. We invoke this result in Section 3.3 as one of several anchors for the non-monotonic fragility hypothesis.

Taschereau-Dumouchel (2025) sharpens this picture with a paradox of centrality. Highly connected hub firms are typically more resilient to idiosyncratic shocks but trigger disproportionately larger cascades of shutdowns when they eventually fail. This asymmetry is the production-network analog of the financial-network non-monotonicity result and directly motivates Supplier HHI as a leading indicator of latent left-tail risk.

The non-monotonic-in-connectivity property formalized in this paper’s predictions is most cleanly stated by Acemoglu et al. (2015). They prove for financial networks that denser interbank connections enhance stability against small shocks but amplify large ones beyond a threshold. Equilibrium networks are socially inefficient because firms do not internalize the systemic externality of their connections. This is the formal anchor for the non-monotonic

fragility framing of the present paper, and the network-externality result explains why concentration arises in equilibrium despite its tail-risk implications.

Acemoglu and Tahbaz-Salehi (2025) establish a closely related result in production networks, proving that equilibrium supply chains are inefficient and that small underlying shocks can generate discontinuous changes in output via firm-level extensive-margin decisions. The mechanism in their paper is a hold-up externality in endogenous network formation rather than a financial-network-style phase transition with a critical shock threshold. The qualitative implication is the same. Small shocks at the wrong nodes produce discontinuous, network-wide damage. We use this paper because it provides micro-founded discontinuous fragility in a production setting with an explicit extensive margin of relationship formation and severance.

Acemoglu et al. (2017) prove that the interaction of idiosyncratic microeconomic shocks with sectoral heterogeneity in Domar weights generates systematic departures in the likelihood of large downturns relative to the normal benchmark, while leaving the central limit theorem intact in the bulk of the GDP distribution. They also show that structural changes can simultaneously reduce aggregate volatility and increase macroeconomic tail risk. This separation between the variance moment and the tail-pricing moment is what motivates the choice of the 25-delta risk reversal, rather than realized variance, as the dependent variable in Section 4.

2.2 Real Dependency and Empirical Proxies

While theoretical models describe the mechanics of network fragility, translating these physical bottlenecks into observable financial signals requires tractable empirical proxies. The non-linearity result of Baqaee and Farhi (2019) establishes the central reason high-frequency proxies are needed at all. Because the predicted effect of concentration on tail risk is non-linear in σ and in the level of concentration, an empirical exercise that conditions on a few well-chosen cross-sectional measures can recover the structural channel without requiring the

analyst to map the full N -tier supply network. Kopytov et al. (2024) formalize a complementary uncertainty channel. They model endogenous network formation when firms face uncertainty over their direct suppliers’ productivity and reroute sourcing toward more stable counterparties even at the cost of higher prices. We use this paper because it captures equilibrium responses to network risk without requiring a behavioral-inattention assumption, which slots cleanly into the discussion of alternative interpretive channels in Section 3.3.

To empirically test this Real Dependency framework, the paper uses market concentration as a high-frequency proxy for structural network bottlenecks. We split this proxy into two distinct dimensions to capture both internal capacity constraints and vertical network topology. First, **Sector HHI** proxies for horizontal, internal consolidation, signaling when an industry’s own macroeconomic footprint (Domar weight) is dangerously concentrated in a few highly complementary nodes.

Sector HHI alone, however, does not account for the vulnerabilities an industry inherits from its critical inputs. We therefore introduce **Supplier HHI**, weighted by Input-Output tables, to proxy for inherited network fragility. Combining Gabaix’s (2011) Granular Hypothesis with Baqaee and Farhi’s (2019) network mechanics, a high Supplier HHI implies that an entire downstream sector relies on a highly concentrated upstream hub. This creates a physical O-ring bottleneck (Kremer, 1993) in which the failure of a single un-substitutable upstream input brings the entire downstream production chain to a halt. We invoke Kremer’s framework because it captures the multiplicative, single-point-of-failure logic that distinguishes Supplier HHI from a generic Herfindahl index of internal market structure.

2.3 Pricing Tail Risk in Financial Markets

Having established Sector and Supplier HHI as observable proxies for horizontal and vertical bottlenecks, we now turn to how financial markets incorporate this latent fragility into asset prices.

The asset pricing literature confirms that both tail risk and network topology are distinct,

priced factors. The formal link between option prices and higher moments of the return distribution comes from Bakshi et al. (2003), who derive a model-free decomposition showing that the prices of out-of-the-money options embed the risk-neutral variance, skewness, and kurtosis of the underlying asset. Their identity provides the theoretical justification for using the implied volatility skew as a direct measure of the market’s forward-looking assessment of crash risk.

Building on this foundation, Kozhan et al. (2013) develop a model-free framework for measuring moment risk premiums and document that the skew risk premium accounts for roughly 40% of the slope of the S&P 500 implied volatility curve. Their finding establishes that the implied volatility skew is a measurable, economically meaningful object whose pricing dynamics can be isolated from those of variance, even though they show that skew and variance premia ultimately load on a common underlying risk factor. The 25-delta risk reversal therefore captures a separable moment-specific signal in the implied volatility surface, regardless of whether its compensation is structurally tied to that of the variance moment.

Hou and Robinson (2006) and Bustamante and Donangelo (2017) establish that the firm-share Herfindahl is a priced characteristic in the cross-section of expected stock returns, operating through markup, entry-threat, and risk-feedback channels. The contribution of the present paper is to extend this concentration-pricing channel from average expected returns to the asymmetric tail-pricing moment, where the predicted relationship is non-monotonic rather than uniform.

Herskovic (2018) bridges production networks and asset pricing. He proves that shocks to the input-output network structure represent an undiversifiable, priced systematic risk factor within the Stochastic Discount Factor. Network connectivity is therefore not merely a feature of the real economy but a state variable that enters the cross-section of expected returns. The empirical exercise in this paper builds on that insight, treating sector- and supplier-level concentration as high-frequency signals about the conditional dynamics of the priced network

factor. Our prediction is directed at the asymmetric tail-pricing moment rather than the average return cross-section. Herskovic is the central bridge between the production-networks literature and the asset-pricing literature, and our introduction leans on this bridge when motivating why the options market should respond to structural concentration at all.

This paper extends the network-pricing result of Herskovic (2018) from general variance to directional tail risk. Under the non-monotonic fragility framing developed in Section 3, the predicted relationship between concentration and the forward risk reversal need not have a single sign across sectors and regimes. At low to moderate concentration, hub formation can compress ordinary-shock variance, which the options market correctly prices with a less negative risk reversal. Beyond a threshold, the same hub structure concentrates tail risk, and where the substitutability parameter σ is low, the tail-pricing branch dominates. Whether the variance-compression branch or the tail-amplification branch shows up in a given sector therefore depends on the effective elasticity of substitution in that sector’s input bundle. That is the empirical content the sector-heterogeneous specifications are designed to recover.

The current literature often separates the real economic mechanics of supply chain bottlenecks from the financial pricing of tail risk. Having established Sector and Supplier HHI as high-frequency proxies for these structural vulnerabilities, this paper bridges that gap. We test whether rising concentration in critical, high-Domar-weight sectors leads the repricing of tail risk, asking whether the options market correctly anticipates the structural cliffs and O-ring bottlenecks created by real economic dependencies, or whether this latent fragility is priced with a lag.

3 Theoretical Foundation

Building on the literature that links network topology to macroeconomic fragility, this section formalizes the transmission mechanism between real supply chain bottlenecks and financial tail risk. The framework first establishes how endogenous network optimization generates

physical skewness in aggregate output, then characterizes the resulting fragility as non-monotonic in concentration, and finally derives the implications for the cross-sectional pricing of forward tail-risk asymmetry in the options market.

3.1 The Formation of Real Dependency and Structural Fragility

This subsection draws on the network-formation arguments of Elliott et al. (2022) and Acemoglu et al. (2012). The foundation of our model is the formation of Real Dependency. Firms do not choose to be fragile; they choose to be efficient. Downstream firms maximize productivity by forming specific, high-quality sourcing relationships and investing in their durability. Elliott et al. (2022) show that the equilibrium of this investment problem can leave the supply network in a regime where small aggregate shocks produce discontinuously large drops in output, even when individual firms multisource as a hedge. The same productivity motive can drive the topology toward the hub-and-spoke configuration that Acemoglu et al. (2012) treat as the canonical asymmetric-network case, in which a small number of upstream nodes account for a disproportionate share of input supply. Once such a relationship is formed, the short-run inability of a new supplier to enter, driven by the fixed costs of qualification and the time required to validate input quality, locks the downstream firm into the dependency until the long run can resolve it.

To make this mechanism concrete, consider a downstream firm j choosing input quantities x_{ij} from a candidate set of suppliers $\{i\}$ to minimize cost subject to producing $\bar{y}_j$ units of output. Suppose first that only one supplier is available. The firm's problem reduces to scaling x_{1j} to meet $\bar{y}_j$, and total cost equals the supplier's marginal cost times the required quantity. Suppose instead that the firm has access to a mix of suppliers, with supplier-specific quality weights $\alpha_{ij} > 0$ that scale the productive contribution of each input. The firm's cost-minimization problem under a CES production technology is to choose $\{x_{ij}\}$ to

minimize $\sum_i p_i x_{ij}$ subject to the production constraint

$$y_j = \left(\sum_i \alpha_{ij}^{1/\sigma} x_{ij}^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)} \geq \bar{y}_j, \quad (1)$$

where p_i is the price charged by supplier i , $x_{ij} \geq 0$ is the quantity sourced from supplier i , $\alpha_{ij} > 0$ is a supplier-specific quality or efficiency weight, and $\sigma > 0$ is the elasticity of substitution governing how readily the firm can swap between suppliers.¹ As $\sigma \rightarrow 0$, the inputs become Leontief and the firm requires every supplier in fixed proportions, so cost minimization yields a smooth interior allocation. As σ rises, the marginal substitutability between suppliers grows and the firm reallocates spending toward the supplier with the highest quality-adjusted productivity. In the limit of heterogeneous quality with sufficiently high σ , standard cost-minimization yields a corner solution in which the firm concentrates all purchases on the single highest-quality supplier. In equilibrium, only one active sourcing relationship is maintained.

The transition from a robust balanced network to a fragile star network fundamentally alters the operative value of σ . In the long run, downstream firms actively optimize for efficiency by abandoning redundant suppliers in favor of the single, highly productive hub with the highest α_{ij} . This corner solution has a critical short-run consequence. With no active alternative supplier, the effective elasticity of the production bundle collapses toward zero, so $\sigma_{\text{eff}} \rightarrow 0$, even when the underlying structural parameter σ is positive. The capacity for substitution exists in principle but has been voluntarily deactivated through endogenous network formation. The fixed costs of qualifying a new supplier and the lead time required to validate input quality prevent rapid reactivation when a hub fails. The endogenous formation of Real Dependency is therefore the precise mechanism that transforms previously substitutable inputs into strict complements over the relevant short-run horizon, setting the

¹Perfect substitutes ($\sigma \rightarrow \infty$) and perfect complements ($\sigma \rightarrow 0$) are limiting cases of the CES specification. The substantive economic content of equation (1) lies in the intermediate region, where a finite σ governs how quickly substitution capacity collapses as concentration rises.

physical stage for the non-linear amplification of downside risk.

3.2 The Physics of Downside Risk and Endogenous Skewness

The structural fragility of the star network materializes through the *extensive margin* of firm operations. Building on Taschereau-Dumouchel (2025), when a key hub fails, downstream firms face a discrete, binary shutdown decision. Production halts entirely rather than declining continuously, because the missing input cannot be substituted at any price in the short run.² The endogenous formation of Real Dependency has transformed previously substitutable inputs into strict complements ($\sigma < 1$).

This lack of substitution fundamentally breaks Hulten’s Theorem (Hulten, 1978). The theorem states that the log-linearized elasticity of aggregate output with respect to a sector’s productivity shock equals that sector’s Domar weight,

$$\frac{\partial \log Y}{\partial \log A_i} = \lambda_i, \quad \lambda_i \equiv \frac{p_i y_i}{p_Y Y}, \quad (2)$$

where A_i is the total factor productivity of sector i , λ_i is its Domar weight (nominal sales as a share of nominal GDP $p_Y Y$), and equation (2) holds under competitive equilibrium with constant returns to scale and no binding complementarities. The linear scaling $\partial \log Y / \partial \log A_i = \lambda_i$ implies that positive and negative shocks of equal magnitude have equal and opposite effects on output, yielding a symmetric output distribution. As Baqaee and Farhi (2019) formalize, strict complementarities ($\sigma < 1$) invalidate this linearity. Positive shocks are dampened by other network bottlenecks while negative shocks are aggressively amplified as the primary bottleneck constrains the entire production chain.

This physical asymmetry creates fundamental negative skewness in real aggregate output, even when the underlying idiosyncratic shocks are strictly Gaussian. Let $\epsilon_i \equiv \log A_i -$

²The extensive margin distinguishes this mechanism from a standard reduction in the *intensive margin* (i.e., a continuous scaling down of output). It is the all-or-nothing nature of the operating decision under strict complementarity that generates the asymmetric, discrete downside tail.

$\mathbb{E}[\log A_i]$ denote the mean-zero log-deviation of sector i 's productivity from its steady-state value, and let $\Delta y \equiv \log Y - \mathbb{E}[\log Y]$ denote the log-deviation of aggregate real output. A second-order Taylor expansion of Δy in the vector of sectoral shocks $(\epsilon_1, \dots, \epsilon_N)$ around the steady state yields (Baqaee and Farhi, 2019),

$$\Delta y \approx \underbrace{\sum_i \lambda_i \epsilon_i}_{\text{Hulten (linear)}} + \underbrace{\frac{1}{2} \sum_{i,j} \frac{\partial^2 y}{\partial \epsilon_i \partial \epsilon_j} \epsilon_i \epsilon_j}_{\text{non-linear amplification}}, \quad (3)$$

where the first-order term recovers Hulten's symmetric approximation and the second-order term captures non-linear amplification. For $i \neq j$, the cross-partial $\frac{\partial^2 y}{\partial \epsilon_i \partial \epsilon_j}$ captures how a shock to one sector affects the marginal impact of a shock to another. In a star network these cross-partials are negative under complementarity, since a downstream sector constrained by two failing hubs simultaneously suffers a worse-than-additive loss. This reinforces the negative second-order contribution (Baqaee and Farhi, 2019). For transparency, we focus the following analysis on the economically dominant diagonal terms.

The Domar weights (λ_i) directly link equation (3) to the Herfindahl–Hirschman Index. By definition, $HHI \equiv \sum_i \lambda_i^2$, so the HHI is the ℓ^2 -norm-squared of the Domar weight vector. For i.i.d. shocks with common variance σ_ϵ^2 , the variance of the first-order term is $\text{Var}(\sum_i \lambda_i \epsilon_i) = HHI \cdot \sigma_\epsilon^2$, so higher concentration directly amplifies first-order output volatility (Gabaix, 2011). Large weights on a few firms therefore increase the magnitude of both terms in equation (3).

A self-contained derivation appears in Appendix A.4, adapting the multi-sector framework of Baqaee and Farhi (2019) to a stylized single-tier CES economy. The sign and magnitude of the diagonal curvature is

$$\frac{\partial^2 y}{\partial \epsilon_i^2} = \lambda_i (1 - \lambda_i) (\sigma - 1). \quad (4)$$

When inputs are complements ($\sigma < 1$), the factor $(\sigma - 1)$ is strictly negative. The sec-

ond derivative is therefore strictly negative for all interior Domar weights ($0 < \lambda_i < 1$). High concentration (large λ_i on a few hub sectors, equivalently large $HHI = \sum_i \lambda_i^2$) multiplied by strong complementarity ($\sigma \ll 1$) produces a large, negative second-order term in equation (3). Through this precise mechanism, the production network mechanically manufactures negative skewness from symmetric shocks.

3.3 Non-Monotonic Fragility and the Pricing Implication

The diagonal-curvature result of equation (4) establishes that concentration manufactures negative skewness in aggregate output whenever inputs are complements. Translating that physical asymmetry into a testable prediction for option prices requires two further ingredients. The first is a formal characterization of how the marginal effect of concentration on aggregate fragility changes with the magnitude of the underlying shock, drawn from Acemoglu et al. (2015) and ported to production networks by Acemoglu and Tahbaz-Salehi (2025). The second is the standard mapping from higher moments of the asset return distribution to option-implied volatilities, due to Bakshi et al. (2003).

The first ingredient is most naturally introduced through the phase-transition results of Acemoglu et al. (2015) for financial networks. Their setting consists of n institutions connected through interbank claims. Each institution receives a stochastic shock to its external assets, and the equilibrium payment vector is determined by a clearing condition on bilateral exposures. The headline result is that there exists a critical shock magnitude $\bar{a} > 0$ that separates two qualitatively different regimes. Below the threshold, denser interconnection allows institutions to share the burden of small idiosyncratic disturbances, and the network behaves as a shock absorber. Above the threshold, the same connections become channels of contagion, and the network behaves as a shock amplifier. Stated in stylized partial-derivative form, with $D(\mathbf{a}; d)$ denoting the equilibrium number of defaulted institutions given shock

vector $\mathbf{a}$ and a diversification index d counting counterparties per institution,

$$\frac{\partial \mathbb{E}[D(\mathbf{a}; d)]}{\partial d} \begin{cases} \leq 0, & \|\mathbf{a}\| < \bar{a}, \\ \geq 0, & \|\mathbf{a}\| > \bar{a}. \end{cases} \quad (5)$$

This is a continuous-index rendering of the comparisons Acemoglu et al. (2015) establish between specific network architectures (complete, ring, and intermediate-density) at small and large shock magnitudes. The paper further shows that equilibrium d is generally socially inefficient, since each institution chooses its connections to maximize its own expected return without internalizing the contribution those links make to systemic risk. Hub formation in our setting is the production-network analog of this externality. Each downstream firm internalizes the private benefit of pairing with the highest-quality supplier but does not internalize the social cost of the concentrated topology that emerges when the same pairing decision is replicated across the economy.

The closest production-network analog of the Acemoglu et al. (2015) phase transition appears in Acemoglu et al. (2017). They prove that the interplay of idiosyncratic microeconomic shocks with sectoral heterogeneity in Domar weights generates systematic departures in the likelihood of large economic downturns relative to what the normal distribution would imply. Crucially for our purposes, they establish that these departures can emerge while GDP fluctuations remain approximately normal away from the tails, and that structural changes in an economy can simultaneously reduce aggregate volatility while increasing macroeconomic tail risks. The mechanism in their paper is that input-output linkages combined with sufficient sectoral dominance translate the tail behavior of micro shocks into asymmetric aggregate tail risk, while leaving the central limit theorem intact in the bulk of the GDP distribution. This is the production-network result that most directly motivates the empirical content of the present paper, since it establishes that concentration affects the tail-pricing moment differently from the variance moment. Acemoglu and Tahbaz-Salehi (2025) provide a com-

plementary mechanism in which equilibrium supply chains exhibit discontinuous responses to small shocks via firm-level extensive-margin decisions, with the discontinuities concentrated at points where the equilibrium network reorganizes. Elliott et al. (2022) establish that the same productivity-investment problem can leave the network in a fragile regime where small shocks generate disproportionately large output drops. Taschereau-Dumouchel (2025) adds that hub firms are more resilient to ordinary shocks but trigger larger cascades when they fail.

These results jointly support an empirical working hypothesis. We expect the production-network concentration measure HHI to govern the conditional behavior of aggregate output deviation Δy in a manner that mirrors the financial-network phase transition of equation (5). The structural hypothesis underlying the empirical work is therefore

$$\frac{\partial}{\partial HHI} \mathbb{E}[\Delta y \mid \|\epsilon\|] \begin{cases} \geq 0, & \|\epsilon\| < \bar{\epsilon} \quad (\text{stabilizing branch}), \\ < 0, & \|\epsilon\| > \bar{\epsilon} \quad (\text{amplifying branch}), \end{cases} \quad (6)$$

where the threshold $\bar{\epsilon}$ separates a regime in which rising concentration absorbs ordinary fluctuations efficiently from one in which the same concentration becomes a propagation channel for extreme shocks. Equation (6) is not a formal theorem. No single existing paper proves it in this exact form for production networks. It is the empirical structure the predictive specifications in Section 4 are designed to detect. Five results jointly motivate it. They are the financial-network phase transition of Acemoglu et al. (2015), the production-network tail-risk result of Acemoglu et al. (2017), the discontinuity result of Acemoglu and Tahbaz-Salehi (2025), the fragile-regime result of Elliott et al. (2022), and the hub-cascade result of Taschereau-Dumouchel (2025). Below the threshold, rising concentration allows hub sectors to absorb routine fluctuations more efficiently than a balanced network of lower-quality alternatives, and the conditional mean output deviation moves toward zero as HHI rises. Above the threshold, the same hub topology becomes a propagation channel that

magnifies extreme shocks through the diagonal-curvature mechanism of equation (4), and the conditional mean output deviation becomes more negative as HHI rises. Concentration and fragility are linked by a curve rather than a line, with a stabilizing branch at moderate HHI and an amplifying branch beyond the threshold.

Equation (6) characterizes the conditional mean response of aggregate output to shocks of different magnitudes, but pricing the asymmetric tail requires the full shape of the aggregate output distribution. The diagonal-curvature result of Section 3.2 supplies that shape. Under complementarity ($\sigma < 1$), equation (4) forces a strictly negative second-order term in the Taylor expansion of Δy , with magnitude scaling in $\lambda_i(1 - \lambda_i)$ and hence concentrated on hub sectors with large Domar weights. The two pieces fit together as follows. The negative second-order term skews the distribution of aggregate output to the left even from symmetric primitive shocks, and the phase transition of equation (6) controls how this skew interacts with concentration. In the stabilizing regime, rising HHI pulls mass toward zero in the body of the distribution, manifesting as variance compression. In the amplifying regime, rising HHI extends the left tail, manifesting as skew amplification.

The mapping from this aggregate output distribution to option prices is mechanical. Bakshi et al. (2003) derive a model-free decomposition that expresses the prices of out-of-the-money options in terms of the risk-neutral variance, skewness, and kurtosis of the underlying asset. The 25-delta risk reversal, defined as the difference in implied volatility between out-of-the-money puts and calls at matched deltas, isolates the third moment from the second. Kozhan et al. (2013) show that the skew risk premium so isolated is a separately priced object that explains roughly half of the slope of the implied volatility curve. Network shocks propagate through the CES production structure and are undiversifiable at the aggregate level (Acemoglu et al., 2012), so the network factor enters the Stochastic Discount Factor (Herskovic, 2018), which closes the loop from the structural concentration measure to the cross-section of option prices. Combining equation (6), the curvature mechanism of equation (4), and the Bakshi et al. (2003) moment decomposition yields the testable

conditional pricing relationship,

$$RR_{j,t+1} = g(HHI_{j,t}; \sigma_j) + \boldsymbol{\gamma}' \mathbf{X}_{j,t} + \varepsilon_{j,t+1}, \quad (7)$$

where $RR_{j,t+1}$ is the 25-delta forward risk reversal for sector j , $g(\cdot)$ is the structural mapping from current concentration to forward tail-risk pricing, σ_j is the effective elasticity of substitution governing the sector's input bundle, $\mathbf{X}_{j,t}$ is a vector of standard asset-pricing controls, and $\varepsilon_{j,t+1}$ is a mean-zero residual. The function g inherits both branches of equation (6). In the stabilizing regime, where $HHI_{j,t}$ is moderate and σ_j is close to one, the variance-compression channel dominates, and the options market correctly prices reduced ordinary-shock volatility through a less negative risk reversal. In the amplifying regime, where $HHI_{j,t}$ is high and σ_j is well below one, the skew-amplification channel dominates, and the risk reversal becomes more negative as concentration rises. The substitutability parameter σ_j governs the location of the boundary between the two regimes. Stronger complementarity shifts weight toward the amplifying branch, weaker complementarity shifts weight toward the stabilizing branch. The sign of $\partial g / \partial HHI_{j,t}$ therefore depends jointly on the level of concentration and on the effective elasticity of substitution in sector j , and the relationship between concentration and the forward risk reversal need not have a single sign across sectors and regimes.³ Section 4 introduces the empirical specifications that recover this dependence by combining two operational concentration measures with sector-heterogeneous coefficient structures.

³An alternative interpretation of any threshold-like patterns in the data assumes rationally inattentive investors with finite information-processing capacity (Sims, 2003), who underreact to gradual changes in concentration until a sufficiently large signal forces a discrete repricing event. Cohen and Frazzini (2008) provide the canonical empirical analog through the predictability of customer-supplier shock transmission. Kopytov et al. (2024) formalize a related uncertainty channel without imposing inattention, modeling endogenous link formation when firms face uncertainty over their direct suppliers' productivity. The rational-inattention channel and the network-non-linearity channel developed in this section are difficult to distinguish on the available data, since both predict threshold-like repricing patterns. The empirical exercise does not require the inattention channel to interpret its findings, and we adopt the network-non-linearity channel as the structural baseline.

4 Data & Methodology

The empirical design must capture both *horizontal* capacity constraints within sectors and *vertical* fragility inherited through the upstream supply chain. Standard sector-level concentration metrics capture internal consolidation but say nothing about the vulnerabilities an industry inherits from its critical inputs. Our baseline structural model therefore incorporates a secondary proxy, Supplier concentration weighted by Input-Output tables, as an observable measure of inherited network fragility. This lets us test whether the options market prices the non-linear tail risks created when an entire downstream sector depends on a highly concentrated upstream hub (Baqae and Farhi, 2019; Acemoglu et al., 2012).

We use a panel predictive regression framework with two-way fixed effects to assess whether the physical structure of market concentration leads the forward-looking financial pricing of tail risk. The staged modeling approach proceeds from a baseline to increasingly granular specifications that allow for sector-level heterogeneity.

4.1 Data Sources and Sample Construction

The study uses a monthly panel dataset built to observe the relationship between sector concentration, upstream supply chain fragility, and the pricing of tail risk. To isolate sector-specific supply chain bottlenecks from broad market sentiment, the universe centers on the eleven Select Sector SPDR ETFs, which partition the S&P 500 into discrete Global Industry Classification Standard (GICS) sectors. The eleven are Technology (XLK), Health Care (XLV), Energy (XLE), Financials (XLF), Industrials (XLI), Consumer Staples (XLP), Consumer Discretionary (XLY), Utilities (XLU), Materials (XLB), Real Estate (XLRE), and Communication (XLC).

The sample window runs from January 2005 through February 2026 across the Bloomberg, BEA, and FRED sources, covering multiple distinct volatility regimes including the Global Financial Crisis, the COVID-19 shock, and the recent technology concentration boom. The

OptionMetrics download terminates on August 29, 2025, and after the one-period forward shift of the 25-delta risk reversal the realized in-sample panel covers 247 months ending in July 2025.

1. **Bloomberg Terminal** (Desktop API): For constituent-level market capitalization, total return indices, and 30-day realized volatility on heavyweight sector constituents.
2. **Bureau of Economic Analysis** (BEA): For annual Make and Use Input-Output tables.
3. **Federal Reserve Economic Data** (FRED): For macroeconomic control variables.
4. **OptionMetrics IvyDB** (via WRDS): For standardized 30-day implied volatility surfaces at the 25-delta and 50-delta marks.

To align the mixed-frequency, multi-source data, all daily identifiers are converted into absolute standard monthly periods. Because certain sector ETFs were launched after the sample start date (Real Estate in 2015, Communication in 2018), the early-period observations for these sectors are kept as missing values. The econometric models therefore use an unbalanced panel design to maximize degrees of freedom without introducing look-ahead bias into market classifications. The final merged panel contains 2,794 sector-month observations, of which 2,425 have complete data on the dependent variable and core controls, spanning 247 months across 11 entities.

4.2 Variable Definitions

4.2.1 Dependent Variable: 25-Delta Risk Reversal

The primary dependent variable is the one-month-ahead 25-delta Implied Volatility Risk Reversal ($RR_{j,t+1}$), which measures the market’s forward-looking expectation of asymmetric tail risk. Following Bakshi et al. (2003), this metric captures the premium investors place on out-of-the-money (OTM) downside protection relative to upside speculation.

Using OptionMetrics IvyDB (via WRDS), we extract standardized implied volatility data for each sector ETF. The WRDS download is pre-filtered to retain only contracts with a 30-day constant maturity, reflecting the standard institutional horizon for hedging near-term tail risk. The resulting dataset contains option-level records with fields for implied volatility, delta, call/put flag, and security identifier. For each sector ETF at the end of each month, we isolate the last available trading day’s option chain and dynamically identify the put and call contracts trading closest to the 25-delta mark. The risk reversal is then computed as

$$RR_{j,t} = IV_{\text{Put}25,t} - IV_{\text{Call}25,t} \quad (8)$$

A rising $RR_{j,t}$ indicates that market participants are bidding up the cost of crash protection relative to upside participation. Because our theoretical framework predicts that concentration drives *future* repricing events, all specifications regress the forward risk reversal ($RR_{j,t+1}$) on period- t explanatory variables. This shift is implemented by sorting within each sector-ticker group and shifting the dependent variable forward by one period.

We also extract the 50-delta put and call implied volatilities using the same closest-delta matching procedure at the ± 0.50 delta mark, then average them to construct the at-the-money implied volatility, $\text{ATM_IV}_{j,t} = (IV_{\text{Put}50,t} + IV_{\text{Call}50,t})/2$. This series enters the in-sample specifications as a level control for the contemporaneous volatility regime, with the rationale developed in Section 4.2.3.

4.2.2 Independent Variables: Concentration and Hub Shock

To capture the structural latent risk described by the Real Dependency framework and the localized stress events that operate as a secondary additive channel, we construct three primary independent variables.

Internal Sector Concentration ($HHI_{j,t}$). To proxy for internal, horizontal capacity constraints, we calculate a dynamic monthly Herfindahl–Hirschman Index for each sector.

To mitigate survivorship bias, the constituent list for each sector’s underlying S&P 500 sector index (e.g., S5INFT for Technology, S5HLTH for Health Care) is sampled annually using the Bloomberg `INDX.MEMBERS` function with a historical end-date override. The union of all historically observed constituents forms the master ticker universe for each sector, and monthly market capitalization (`CUR.MKT.CAP`) is extracted for all such firms. Firms with missing market capitalization in a given month (e.g., due to delisting) are assigned a value of zero, effectively removing them from the concentration calculation for that period.⁴ HHI is then computed as

$$HHI_{j,t} = \sum_{i=1}^{N_j} \left(\frac{MktCap_{i,t}}{\sum_k MktCap_{k,t}} \right)^2 \quad (9)$$

Inherited Upstream Fragility (*Supplier HHI_{j,t}*). To capture how systemic risk cascades downstream through the production network, we construct a network-weighted concentration metric using annual Make and Use Input-Output tables from the BEA. The raw BEA commodity-by-industry tables are mapped to GICS sector classifications using a manual concordance dictionary that matches BEA industry name strings to the eleven GICS sectors. Industries that do not map to any GICS sector (e.g., government, non-profit services) are classified as “Other” and excluded from the weight matrix.⁵

For every downstream sector j , the proportion of inputs sourced from each upstream sector k is calculated as the weight $\omega_{j,k}$. The matrix diagonal is set to zero to strictly capture inter-sector propagation. To prevent look-ahead bias, the I-O weight matrix is lagged by one year. The Supplier HHI for any month in year t uses the weight matrix derived from the year $t - 1$ tables, with the earliest available year used as a fallback for boundary observations,

$$SupplierHHI_{j,t} = \sum_{k \neq j} \omega_{j,k} HHI_{k,t} \quad (10)$$

⁴This approach avoids forward-filling stale data for potentially delisted firms but may slightly understate true concentration in months where a major constituent experiences a temporary data gap.

⁵The BEA-to-GICS concordance relies on keyword matching against industry description fields. This mapping is optimized for coverage but is an approximation; robustness checks with alternative concordances are left for future work.



Figure 1: Inter-sector input-output weight matrix $\omega_{j,k}$ from the 2024 BEA tables. Cells give the share of downstream sector j 's (column) inputs sourced from upstream sector k (row); the diagonal is zero by construction. Color scale capped at the 95th percentile of off-diagonal values.

Combining Gabaix's (2011) Granular Hypothesis with Baqaee and Farhi's (2019) network mechanics, a high *SupplierHHI* implies that an entire downstream sector relies on a highly concentrated upstream hub. The result is a physical O-ring bottleneck where the failure of a single non-substitutable input can halt the entire downstream production chain. Figure 1 displays the inter-sector input-output weight matrix from the most recent BEA tables. The dominance of Industrials as an upstream supplier across the economy and the Energy-to-Utilities concentration are visible at a glance, providing context for the cross-sector heterogeneity in Supplier HHI documented in Figure 2. The latter figure plots Supplier HHI by sector over the full sample. It shows a secular decline in inherited upstream concentration from 2005 through roughly 2018 followed by a partial reversal coinciding with the COVID-era supply chain disruptions and subsequent re-shoring trends. Sectors for which the Model 3B Supplier HHI sector interaction is statistically significant under entity-clustered standard errors are highlighted in navy.

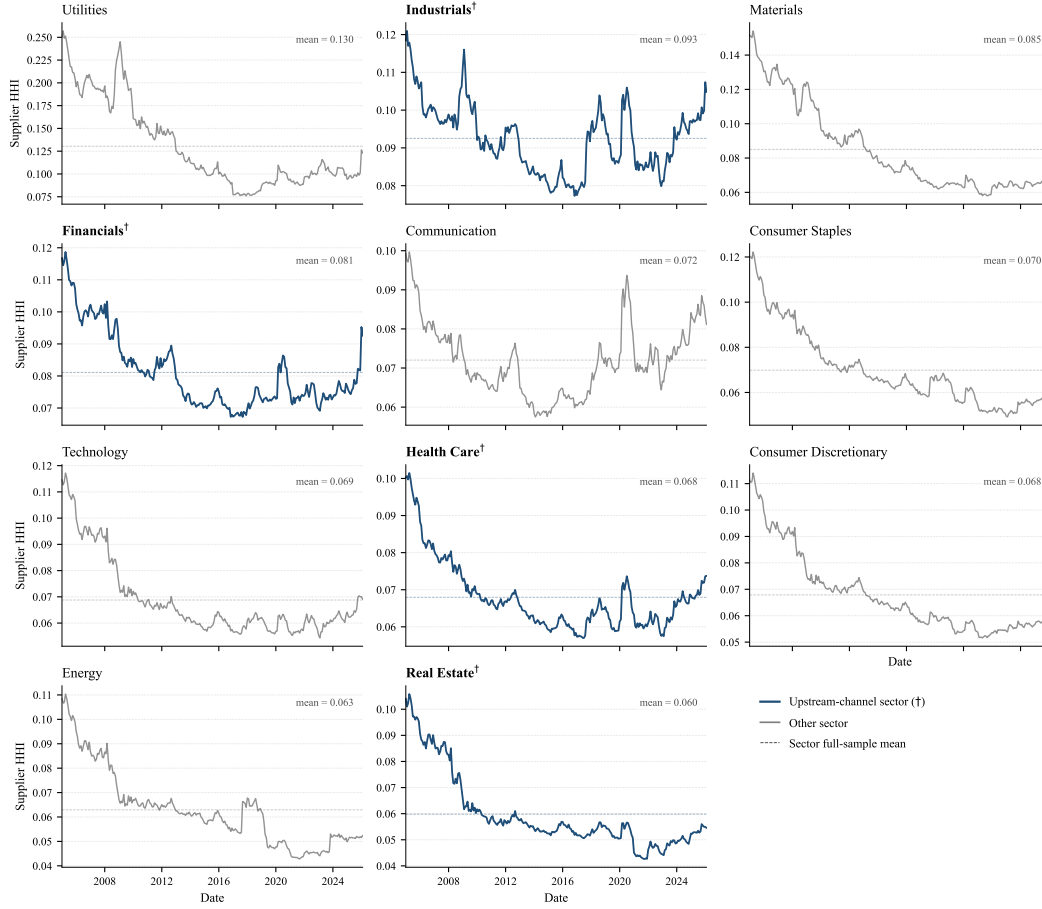


Figure 2: Supplier HHI by sector, 2005–2026. Panels are sorted by full-sample mean and use sector-specific y-axes. The † marker flags sectors whose Model 3B Supplier HHI interaction is significant at the 5% level under entity clustering.

Hub Shock ($HubShock_{j,t}$). To capture localized stress events in heavyweight sector constituents, we construct a binary variable that flags volatility spikes in firms that are currently central to their sector. For each sector and each month t , we first define the heavyweight set as the union of all firms appearing among the three largest constituents by market capitalization in any month within the rolling 36-month window ending at t . The rolling window departs from a full-sample heavyweight set so that the trigger reflects *currently* central nodes rather than firms whose dominance was strictly historical. Monthly 30-day realized volatility (VOLATILITY_30D from Bloomberg, computed on daily price returns) is then extracted for the universe of historically observed heavyweight firms across the full sample. For each firm we compute a rolling 12-month mean and standard deviation of realized volatility. The Hub Shock for sector j in month t equals one if any firm in the rolling-window heavyweight set for that sector experiences a 30-day realized volatility exceeding two standard deviations above its rolling 12-month mean, and zero otherwise.

This construction restricts the monitoring set to firms that are dominant within a recent window rather than firms that were dominant at any point in the full sample. The rationale is that the production-network relationships generating systemic stress are tied to a firm’s contemporaneous position in the sector hierarchy. A volatility spike in a firm whose dominance has lapsed for several years carries less informational content for current network fragility than a spike in a firm that is currently among the top three.⁶

4.2.3 Control Variables

To isolate the structural network effect from ambient market conditions, we include controls across two dimensions. At the sector level, we include the at-the-money implied volatility ($ATM_IV_{j,t}$) defined in Section 4.2.1 to absorb the contemporaneous volatility regime, and a Bloomberg total return index ($TOT_RETURN_INDEX_GROSS_DVDS$) to control for momentum

⁶An earlier specification used the full-sample union of top-three constituents and produced a coefficient of 0.0011 ($t = 1.97$). The rolling-window construction lifts the coefficient to 0.0014 ($t = 2.53$) under entity clustering, consistent with sharper identification of the hub-shock signal once historically dominant but currently peripheral firms are excluded from the heavyweight set.

effects. At the macro level, we include the CBOE VIX Index (VIXCLS from FRED, resampled to end-of-month frequency), the term spread (10-year minus 2-year U.S. Treasury constant maturity rate), and the credit spread (Moody’s Baa minus Aaa seasoned corporate bond yields). These macro series do not enter the in-sample regressions as direct regressors because the time fixed effects absorb their variation entirely. They appear as direct regressors only in the out-of-sample forecasting specifications, where time fixed effects cannot be projected onto future periods and the macro series substitute for them.

ATM_IV enters as a level measure of the contemporaneous volatility regime, absorbing the broad shifts in implied volatility that move both wings of the option-pricing distribution symmetrically. Controlling for the level of the volatility surface separately from its skew ensures that the structural concentration coefficients are identified from variation in tail-risk asymmetry rather than from co-movement with the overall volatility regime.⁷ Empirically, ATM_IV is insignificant in every in-sample specification, which is the expected behavior for a level control that absorbs ambient volatility without absorbing predictive content.⁸

4.3 Empirical Strategy

The staged econometric approach separates latent structural risk from acute repricing events and tests whether the structural mechanism operates homogeneously or varies across sectors. All in-sample models use two-way fixed effects for sector (α_j) and time (γ_t) and predict the one-month-ahead risk reversal ($RR_{j,t+1}$). Standard errors are clustered at the entity level throughout to account for within-sector serial correlation, and Driscoll-Kraay standard errors with bandwidths of six and twelve months are reported in the appendix as a robustness check on the headline coefficients.

⁷An earlier specification used the 50-delta risk reversal as the volatility-regime control. Since the 50-delta risk reversal is itself a near-ATM skew measure, partialing it out of a regression on the 25-delta risk reversal inflated the in-sample R^2 through mechanical autocorrelation. The Model 1 Supplier HHI t -statistic tightens from -4.96 to -5.22 under the ATM_IV specification.

⁸ATM_IV and the VIX exhibit substantial within-sector collinearity ($\rho = 0.89$, with VIFs of 18.6 and 27.1). The in-sample specifications are unaffected because the time fixed effects absorb the VIX entirely. The out-of-sample specifications drop the VIX and retain ATM_IV to avoid joint inflation of the standard errors, as detailed in Section 4.3.4.

4.3.1 Model 1: Baseline Structural Model

The first specification tests the Real Dependency hypothesis uniformly across the market. It evaluates whether internal concentration and inherited upstream fragility jointly predict a forward-looking premium on downside option protection,

$$RR_{j,t+1} = \alpha_j + \gamma_t + \beta_1 HHI_{j,t} + \beta_2 SupplierHHI_{j,t} + \boldsymbol{\theta}'\mathbf{X}_{j,t} + \varepsilon_{j,t+1} \quad (11)$$

where $\mathbf{X}_{j,t}$ includes $ATM.IV_{j,t}$ and the total return index as sector-level controls. This is the workhorse specification. Including both concentration dimensions in a single regression lets us assess whether horizontal and vertical fragility carry independent predictive content. Under the non-monotonic fragility framework formalized in Section 3.3, a significant negative coefficient on *SupplierHHI* is consistent with the variance-compression branch of the relationship. Rising upstream concentration reduces ordinary-shock variance, and the options market prices that lower variance with a less negative forward risk reversal.

4.3.2 Model 2: Additive Hub Shock Channel

The second specification tests whether localized stress events in heavyweight firms operate as an independent additive channel alongside the slow-moving structural concentration signal. We add *HubShock_{j,t}* as a predictor alongside internal concentration,

$$RR_{j,t+1} = \alpha_j + \gamma_t + \beta_1 HHI_{j,t} + \beta_2 HubShock_{j,t} + \boldsymbol{\theta}'\mathbf{X}_{j,t} + \varepsilon_{j,t+1} \quad (12)$$

An earlier specification interacted *HHI* with *HubShock* to test whether concentration amplifies the repricing effect of localized hub-firm volatility spikes. The interaction term proved statistically insignificant ($t = -0.41$), indicating that the two variables operate through separate channels rather than a joint amplification mechanism. The additive specification reported here therefore tests whether acute distress in hub firms triggers an independent

level shift in next-month’s forward skew. Hub Shock is treated throughout as a secondary additive channel rather than the central transmission mechanism.

4.3.3 Models 3A–3C: Sector Heterogeneity Specifications

The theoretical framework predicts that the elasticity of substitution (σ) varies across sectors. Physical production networks featuring strict complementarities ($\sigma < 1$) should amplify downside shocks more aggressively than digital or service-oriented networks where substitution is easier. To investigate heterogeneous responses to concentration, we estimate three nested specifications that interact our structural metrics with sector indicator variables (D_j). The indicator D_j equals one when an observation belongs to sector j and zero otherwise. Dummies are built using standard drop-first encoding with Communication as the omitted reference category. The summations in equations (13) through (15) therefore run over the ten non-omitted sectors, and each $\beta_{1,j}$ or $\beta_{2,j}$ measures the differential loading on the corresponding structural variable for sector j relative to Communication. To identify the Communication-sector loading itself, the estimation in Models 3A and 3B additionally includes a homogeneous HHI term that is suppressed from the equation display for compactness, and Models 3B and 3C further include a homogeneous $SupplierHHI$ term in the same role. Model 3C displays the corresponding homogeneous HHI baseline explicitly in equation (15).

Model 3A (Heterogeneous HHI , Homogeneous $SupplierHHI$). This specification allows the internal concentration effect to vary by sector while constraining the upstream fragility channel to a uniform coefficient. It isolates whether the *internal* pricing of concentration differs between, say, Technology, where high HHI may reflect platform-economy dynamics with higher substitutability, and Industrials, where it reflects physical bottlenecks with strict

complementarities,

$$RR_{j,t+1} = \alpha_j + \gamma_t + \sum_j \beta_{1,j} (HHI_{j,t} \times D_j) + \beta_2 SupplierHHI_{j,t} + \boldsymbol{\theta}' \mathbf{X}_{j,t} + \varepsilon_{j,t+1} \quad (13)$$

Model 3B (the Kitchen Sink). This fully saturated specification interacts *both* internal and supplier concentration with sector dummies simultaneously. It serves as a diagnostic tool to identify whether sector-level heterogeneity operates primarily through the internal or the upstream channel,

$$RR_{j,t+1} = \alpha_j + \gamma_t + \sum_j \beta_{1,j} (HHI_{j,t} \times D_j) + \sum_j \beta_{2,j} (SupplierHHI_{j,t} \times D_j) + \boldsymbol{\theta}' \mathbf{X}_{j,t} + \varepsilon_{j,t+1} \quad (14)$$

Model 3C (the Alternating model). This specification reverses the heterogeneity structure. Internal concentration enters with a homogeneous baseline coefficient while supplier concentration is interacted with sector dummies. The specification directly tests the Supplier Paradox, asking whether the options market prices upstream technology monopolies differently from upstream industrial monopolies,

$$RR_{j,t+1} = \alpha_j + \gamma_t + \beta_1 HHI_{j,t} + \sum_j \beta_{2,j} (SupplierHHI_{j,t} \times D_j) + \boldsymbol{\theta}' \mathbf{X}_{j,t} + \varepsilon_{j,t+1} \quad (15)$$

The progression from Model 3A through 3C serves a diagnostic function. Comparing coefficient patterns across these nested specifications lets us determine whether sector heterogeneity in tail-risk pricing originates from differences in internal market structure, from differences in the composition and fragility of upstream supply chains, or from both channels at once.

4.3.4 Out-of-Sample Forecasting

To guard against in-sample overfitting, we complement the fixed-effects inference models with an out-of-sample (OOS) forecasting exercise. Because two-way time fixed effects cannot be projected onto future periods, the forecasting specifications substitute the time effects with the macroeconomic time-series controls described in Section 4.2.3, alongside the sector-specific controls and entity fixed effects.

The sample is partitioned into a *training* window (January 2005 through December 2019) and a *test* window (January 2025 through February 2026). The intervening period (2020–2024) is excluded from both windows. Because the OptionMetrics download terminates on August 29, 2025 and the one-period forward shift truncates the final observation, the realized test sample contains seven months of forecasts (January through July 2025) per sector ETF, yielding 77 forecast observations across the eleven sectors.

This design insulates the training sample from the extreme volatility regimes of the COVID-19 pandemic and its monetary-policy aftermath, while testing the model’s predictive content on a genuinely out-of-distribution evaluation set drawn from the most recent data.⁹

The OOS forecasting controls are ATM_IV, the total return index, the term spread, and the credit spread, with entity fixed effects retained throughout. The VIX is dropped from the OOS specifications because of its high within-sector correlation with ATM_IV ($\rho = 0.89$), and the in-sample identification is unaffected because time fixed effects absorb the VIX in those models.

We estimate three structural challenger models on the training sample and generate predictions on the held-out test set. The *baseline* model includes only the macroeconomic and sector-specific controls listed above. The *homogeneous challenger* augments the baseline with *HHI* and *SupplierHHI*. The *heterogeneous challenger* corresponds to Model 3A

⁹An earlier expanding-window OOS exercise was attempted with an initial training cutoff of December 2014, but the late entry of XLRE (2015) and XLC (2018) into the sector universe generated more than forty absorbing-effect failures. The expanding-window specification was therefore dropped in favor of the fixed-window design reported here.

and augments the baseline with sector-interacted HHI coefficients alongside homogeneous $SupplierHHI$, addressing the concern that homogeneous specifications may understate the true predictive content of concentration when the underlying mechanism varies by sector. The third challenger is the fully saturated kitchen-sink specification (Model 3B). Forecast accuracy is compared via the Mean Squared Prediction Error (MSPE),

$$MSPE = \frac{1}{T} \sum_t \left(RR_{j,t+1} - \widehat{RR}_{j,t+1} \right)^2 \quad (16)$$

and the statistical significance of the forecast-error differentials is assessed via a Diebold-Mariano test on the loss-difference series. If the structural challengers achieve a meaningful reduction in MSPE relative to the baseline, this constitutes evidence that concentration metrics contain genuine predictive content for tail-risk pricing beyond what is captured by standard market-condition variables. The interpretation of the OOS results is developed in Section 5.4 against the backdrop of the in-sample inference.

4.4 Identification, Robustness, and Estimation Notes

Several features of the empirical design are worth flagging. First, the forward-looking structure of the dependent variable ($RR_{j,t+1}$ regressed on period- t regressors) ensures that any documented relationship is genuinely predictive rather than contemporaneous. Second, the two-way fixed-effects structure absorbs time-invariant sector characteristics (α_j) and common aggregate shocks (γ_t), isolating the within-sector, within-period variation in concentration that drives tail-risk repricing. Third, standard errors clustered at the entity level account for serial correlation within each sector’s time series, and Driscoll-Kraay standard errors at bandwidths of six and twelve months are reported as a robustness check on the headline coefficients. Fourth, the unbalanced panel design avoids introducing artificial observations for recently launched sector ETFs, ensuring that all estimation relies on genuinely observed market data.

A note on the use of contemporaneous rather than lagged *HHI* is warranted. Under the non-monotonic fragility framework developed in Section 3.3, the structural relationship between concentration and fragility is a property of the current network state, not a result of investors processing information slowly. Contemporaneous *HHI* therefore captures the present structural condition that determines current and forward fragility, and the forward-shifted dependent variable ensures that the documented relationship is predictive rather than coincident. A distributed-lag specification of *HHI* would be the appropriate design under an alternative theoretical framing in which markets reprice fragility with a delay, but that framing is not the one this paper adopts.

The robustness suite reported in the appendix complements the main specifications along two further dimensions. The Driscoll-Kraay standard errors at bandwidths of six and twelve months relax the within-entity clustering assumption and accommodate cross-sectional dependence that can arise in production networks. A leave-one-year-out cross-validation exercise within the in-sample training window (fifteen folds covering 2005–2019) reports the fold-to-fold sign stability and significance frequency of headline coefficients, providing a second perspective on which results are robust and which are bandwidth-dependent. The two robustness exercises align in identifying the Model 1 Supplier HHI coefficient and the Model 3A internal-HHI substitutability gradient as robust headline findings. The Model 3B and Model 3C upstream-channel sector interactions lose 5% significance under DK6 and exhibit fold-to-fold instability under leave-one-year-out cross-validation. We treat the latter as exploratory diagnostics for identifying which sectors plausibly carry the upstream channel rather than as confirmatory point estimates.

Two data construction choices warrant transparency. The BEA-to-GICS concordance relies on a manual keyword-matching dictionary that maps BEA commodity industry names to GICS sectors. We have optimized this mapping for coverage, but some BEA industries (e.g., government, non-profit services) do not correspond to any traded GICS sector and are excluded. The I-O weight matrix therefore captures inter-sector flows among traded

industries only. Additionally, firms with missing Bloomberg market capitalization data in a given month are assigned a zero value, effectively dropping them from the HHI calculation for that period, as noted in Section 4.2.2.

5 Results & Discussion

This section reports the results of the staged econometric analysis described in Section 4. We begin with the baseline structural model, proceed through the additive Hub Shock specification and the sector heterogeneity models, and conclude with the out-of-sample forecasting exercise. All in-sample models use two-way (entity and time) fixed effects with standard errors clustered at the entity level. Driscoll-Kraay standard errors with bandwidths of six and twelve months are reported in Appendix B as a robustness check. Tables 1 and 4 report coefficient estimates and t -statistics for the full set of in-sample specifications, Table 5 reports the out-of-sample comparison, and Tables 2 and 3 translate the Model 3A interaction estimates into net sector-specific effects and economic-magnitude basis-point equivalents.

5.1 Model 1: Baseline Structural Model

The baseline structural model in equation (11) provides the cleanest test of how horizontal and vertical concentration jointly predict the forward pricing of tail risk. Of the two concentration dimensions, Supplier HHI is the dominant predictor of the one-month-ahead 25-delta risk reversal, with a coefficient of -0.0758 ($t = -5.22$, $p < 0.001$). Internal sector concentration, by contrast, is economically small and statistically indistinguishable from zero ($\hat{\beta}_{\text{HHI}} = -0.0073$, $t = -1.06$). The model is jointly significant at the 1% level ($F = 4.07$, $p = 0.003$).

The negative sign on Supplier HHI is consistent with the variance-compression branch of the non-monotonic fragility mechanism formalized in Section 3.3. Under the Acemoglu et al. (2015) phase-transition logic, ported to production networks by Acemoglu and Tahbaz-

Table 1: Main In-Sample Results: Forward Risk Reversal Predictive Regressions

	(1)	(2)	(3)
	M1: Baseline	M2: Hub Shock	M3A: Het. HHI
<i>Panel A. Homogeneous structural predictors</i>			
HHI	−0.0073 (−1.06)	0.0023 (0.21)	−0.0859*** (−4.94)
Supplier HHI	−0.0758*** (−5.22)		−0.0632*** (−4.65)
Hub Shock		0.0014** (2.53)	
<i>Panel B. Controls</i>			
ATM IV	0.0032 (0.40)	0.0042 (0.55)	0.0056 (0.65)
Total Return Index	4.41×10^{-5} *** (2.91)	2.31×10^{-5} (1.29)	5.04×10^{-5} ** (2.40)
Sector $\times$ HHI (M3A)	—	—	See Table 2
Entity FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Observations	2,425	2,425	2,425
R^2 (within)	−0.052	−0.018	−0.035
F -statistic	4.07	1.50	1.82
F p -value	0.003	0.200	0.031

Dependent variable: $RR_{j,t+1}$, the one-month-ahead 25-delta IV risk reversal. t -statistics in parentheses, entity-clustered. M3A includes ten HHI $\times$ sector interactions; Communication is the omitted reference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Salehi (2025), the marginal effect of upstream concentration on aggregate fragility need not be monotone. At low to moderate concentration, hub formation reduces vulnerability to ordinary idiosyncratic shocks because central nodes diversify across many users and absorb fluctuations efficiently. The options market reads this lower ordinary-shock variance correctly and prices it with a less negative forward risk reversal. The same hub structure also accumulates latent extreme-tail exposure that may materialize only beyond the threshold shock magnitude $\bar{a}$, but whether the tail repricing dominates the variance compression in any given specification depends on substitutability and on the level of concentration. The homogeneous coefficient in Model 1 is best read as picking up the variance-compression branch in the cross-section average, with the tail-risk branch becoming visible only once sector heterogeneity is permitted in Section 5.3.

Among the controls, the level proxy for the volatility regime, ATM IV, is statistically insignificant ($t = 0.40$). This is the correct behavior for a level control whose role is to soak up sector-time volatility variation rather than predict next-month skew. The total return index is positive and significant in Model 1 ($t = 2.91$), suggesting a modest momentum channel in tail-risk pricing, although this control loses significance in several heterogeneous specifications (Table 1, columns 2 and 3 of Panel B).

5.2 Model 2: The Hub Shock Channel

The second specification in equation (12) tests whether acute volatility events in heavyweight constituents trigger an independent additive catalyst for the repricing of latent structural risk. An earlier formulation interacted internal HHI with Hub Shock to test for joint amplification, but the interaction term was statistically insignificant ($t = -0.41$), and we report the additive specification here.

Hub Shock carries a positive coefficient ($\hat{\beta}_{\text{HubShock}} = 0.0014$). It is significant at the 5% level under entity-clustered standard errors ($t = 2.53$) and marginal under Driscoll-Kraay

with a six-month bandwidth ($t = 1.90$).¹⁰ The coefficient is small in absolute magnitude relative to the structural concentration variables, and the bandwidth sensitivity confirms that Hub Shock should be read as a secondary additive channel rather than the central transmission mechanism. Volatility spikes in currently-central firms generate small upward adjustments in next-month forward tail-risk pricing that operate independently of the slow-moving structural concentration signal. Internal HHI itself remains insignificant in this specification ($t = 0.21$), reinforcing the finding from Model 1 that a homogeneous, market-wide treatment of internal concentration lacks predictive power.

The relative weakness of the Hub Shock channel against the strength of the Supplier HHI effect in Model 1 indicates that the primary mechanism through which concentration shapes forward skew operates through slow-moving structural dependencies in the production network rather than through high-frequency stress events at the firm level. Acute hub-firm distress contributes a level shift, but it does not amplify the structural channel.

5.3 Sector Heterogeneity in Tail-Risk Pricing

The theoretical framework predicts that the mapping from concentration to forward tail risk should vary with the effective elasticity of substitution σ , which differs across sectors depending on the physical nature of their production networks. The diagonal-curvature result in equation (4) delivers a strictly negative second-derivative term whenever $\sigma < 1$, so sectors operating with stronger short-run input complementarities should display larger positive loadings on internal concentration. Sectors closer to the unit-elasticity case should display loadings near zero. The sector heterogeneity specifications confirm this prediction directly, and they show that the homogeneous insignificance of HHI in Models 1 and 2 conceals substantial cross-sector variation rather than reflecting a true zero effect.

¹⁰Both bandwidths are reported in Appendix B. Significance attenuates further at twelve months but remains directionally consistent.

5.3.1 Model 3A: Heterogeneous Internal Concentration

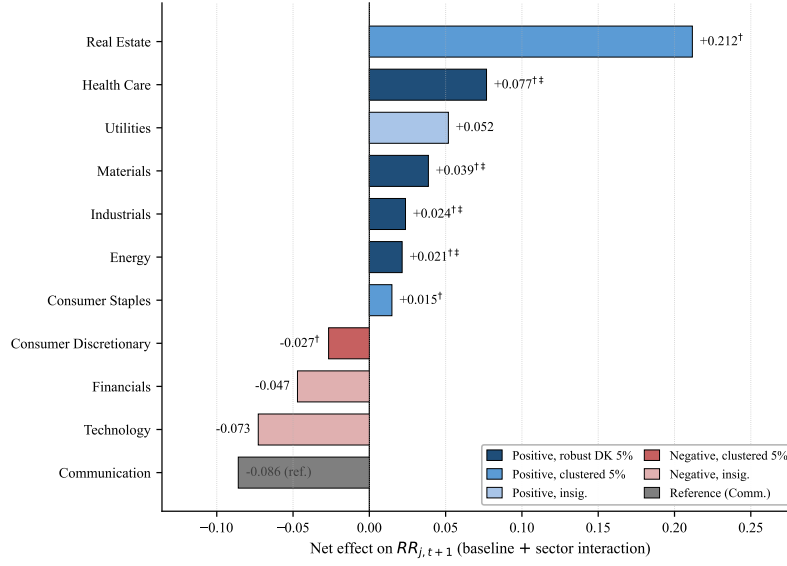


Figure 3: Net sector-specific HHI effects under Model 3A. Markers refer to the interaction term tested against the omitted Communication reference. † marks 5% entity-clustered significance, ‡ marks 5% significance also under Driscoll-Kraay (bw = 6).

Letting the HHI coefficient vary by sector under equation (13) transforms the picture. The baseline HHI coefficient, which captures the loading for the omitted reference sector (Communication), is strongly negative ($\hat{\beta}_{\text{HHI}} = -0.086$, $t = -4.94$). Eight of the ten sector interaction terms are positive, and seven of these reject equality with the Communication baseline at the 1% level under entity-clustered standard errors. Figure 3 displays the resulting net sector-specific effects, with significance markers showing which interactions remain significant under both clustering and Driscoll-Kraay (DK6) standard errors. Tables 2 and 3 report the underlying numbers and translate them into basis-point implications for a one-standard-deviation move in sector HHI.

The net effects reveal a sharp and theoretically interpretable economic gradient. Real Estate stands alone at the top of the distribution with a net effect of +0.212, which translates under Table 3 into a +9.9 basis-point change in the forward 25-delta risk reversal for a one-standard-deviation increase in sector HHI. Six other sectors cluster at smaller but con-

Table 2: Net Sector-Specific HHI Effects on Forward Risk Reversal (Model 3A)

Sector	Baseline (omitted)	Sector interaction	Net effect (sum)	Interaction t (clustered)	Interaction t (DK, bw = 6)
<i>Sectors with positive net effect (consistent with $\sigma < 1$ regime)</i>					
Real Estate	-0.086	+0.298	+0.212	3.52***	0.95
Health Care	-0.086	+0.163	+0.077	3.18***	2.29**
Utilities	-0.086	+0.138	+0.052	1.35	0.67
Materials	-0.086	+0.125	+0.039	4.29***	2.18**
Industrials	-0.086	+0.110	+0.024	5.68***	2.54**
Energy	-0.086	+0.107	+0.021	4.49***	2.94***
Consumer Staples	-0.086	+0.101	+0.015	3.90***	1.90*
<i>Sectors with negative or near-zero net effect (consistent with $\sigma \approx 1$ regime)</i>					
Consumer Discretionary	-0.086	+0.059	-0.027	3.20***	1.60
Financials	-0.086	+0.039	-0.047	0.51	0.23
Technology	-0.086	+0.013	-0.073	0.51	0.24
Communication (ref.)	-0.086	—	-0.086	-4.94***	-2.59***

Net effect equals the homogeneous baseline coefficient ($\hat{\beta}_{HHI} = -0.086$, the loading for the omitted Communication category) plus the sector interaction term. Reported t -statistics test that the sector *interaction* term equals zero, not that the net effect itself does. DK uses a Bartlett kernel, bandwidth six. The Communication row reports the baseline coefficient and its t -statistic directly. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Economic Magnitude of Net Sector-Specific HHI Effects (Model 3A)

Sector	Mean HHI	SD HHI	Net coef. (M3A)	ΔRR per $+1\sigma$ HHI (decimal)	ΔRR per $+1\sigma$ HHI (basis points)
Communication (ref.)	0.172	0.028	-0.086	-0.0024	-23.7
Consumer Discretionary***	0.102	0.050	-0.027	-0.0013	-13.4
Technology	0.115	0.017	-0.073	-0.0012	-12.4
Real Estate***	0.055	0.005	+0.212	+0.0010	+9.9
Energy***	0.176	0.045	+0.021	+0.0010	+9.8
Health Care***	0.053	0.011	+0.077	+0.0009	+8.5
Industrials***	0.052	0.032	+0.024	+0.0007	+7.5
Materials***	0.077	0.016	+0.039	+0.0006	+6.3
Consumer Staples***	0.093	0.022	+0.015	+0.0003	+3.3
Financials	0.048	0.005	-0.047	-0.0002	-2.1
Utilities	0.059	0.004	+0.052	+0.0002	+1.8

Net coefficient equals the M3A baseline (-0.086) plus the sector interaction. ΔRR per $+1\sigma$ HHI is the predicted change in $RR_{j,t+1}$ for a one-standard-deviation rise in sector HHI; basis points equal decimals $\times 10,000$. Significance markers refer to the sector interaction term: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Sorted by absolute basis-point impact.

sistently positive magnitudes. Health Care leads the cluster at +0.077, followed by Utilities at +0.052, Materials at +0.039, Industrials at +0.024, Energy at +0.021, and Consumer Staples at +0.015. These seven sectors share physical-capital intensity, long-lived asset specificity, and production processes in which critical inputs are difficult to substitute on short notice. These are precisely the conditions under which $\sigma < 1$ and the second-order term in equation (3) dominates. The remaining three sectors load negatively. Technology ($-0.086 + 0.013 = -0.073$, interaction $t = 0.51$) and the reference Communication sector (-0.086) operate in production environments where digital inputs can be rerouted and platform infrastructure provides built-in redundancy, dampening the network curvature mechanism. Financials and Consumer Discretionary fall in the middle. Consumer Discretionary’s interaction is significant under clustering ($t = 3.20$) but not robust under DK6 ($t = 1.60$), and Financials’ interaction is insignificant under both ($t_{\text{clust}} = 0.51$, $t_{\text{DK}} = 0.23$).

The clustering of seven physical-capital-intensive sectors above the zero line, with the three digital and platform-economy sectors below, is the headline empirical finding. It is the substitutability gradient predicted by the theory in Section 3, and it is what the homogeneous specifications in Models 1 and 2 average over.

Two cautions on the Real Estate point estimate are worth flagging. First, Real Estate’s interaction coefficient, while strongly significant under entity clustering ($t = 3.52$), attenuates substantially under Driscoll-Kraay ($t = 0.95$). Second, the leave-one-year-out cross-validation summary reported in the appendix shows Real Estate as the least stable interaction across folds, with significance at the 5% level in only 27% of holdout years. Both diagnostics trace to the relatively short within-sector panel for Real Estate, since XLRE entered the sector universe in late 2015 and contributes roughly four fewer years of within-sector training data than the legacy ETFs.¹¹ The Real Estate point estimate should be read as directionally consistent with the substitutability gradient rather than as a precisely identified magnitude. The remaining six positive sectors are robust under both DK6 and

¹¹The DK attenuation is consistent with the residual structure picking up the smaller effective sample for XLRE; see Appendix B for the full DK6/DK12 comparison and the LOO CV stability summary.

LOO. Health Care, Materials, Industrials, and Energy are each significant at the 5% level, and Consumer Staples at the 10% level, under both inference regimes.

When Supplier HHI is reintroduced as a homogeneous control alongside the heterogeneous internal-HHI structure, it retains its strong negative coefficient ($\hat{\beta}_{\text{SuppHHI}} = -0.0632$, $t = -4.65$), and the heterogeneous HHI interaction pattern remains qualitatively unchanged. The joint F -test rejects at the 5% level ($F = 1.82$, $p = 0.031$). The two concentration channels carry independent, non-redundant predictive content for forward tail-risk pricing.

5.3.2 Models 3B and 3C: Disentangling Horizontal and Vertical Heterogeneity

The fully saturated Kitchen Sink specification (Model 3B, equation (14)) interacts both HHI and Supplier HHI with sector indicators simultaneously. The Alternating specification (Model 3C, equation (15)) reverses the heterogeneity structure, holding HHI homogeneous while interacting Supplier HHI with sector dummies. Together these specifications act as exploratory diagnostics for identifying which sectors plausibly carry the upstream channel independently of the within-sector internal-concentration channel. We do not treat the resulting sector-specific point estimates as confirmatory.

Two patterns are worth noting. First, the homogeneous Supplier HHI coefficient collapses to statistical insignificance in M3B ($\hat{\beta}_{\text{SuppHHI}} = 0.028$, $t = 0.57$) once the upstream channel is permitted to vary by sector. The strong negative effect observed in Models 1 and 3A is an average that conceals sharply divergent sector-specific responses. Second, four sectors stand out under M3B clustering as the candidates for which inherited upstream concentration positively predicts forward tail-risk pricing. Real Estate carries the largest interaction ($+0.406$, $t = 4.06$), followed by Financials ($+0.298$, $t = 3.02$), Health Care ($+0.195$, $t = 3.01$), and Industrials ($+0.118$, $t = 3.33$). These four sectors share a common structural feature. Each depends critically on upstream inputs characterized by high asset specificity and low short-run substitutability. Financials draws on concentrated technology and clearing infrastructure, Health Care relies on specialized pharmaceutical and device supply chains,

Table 4: Upstream-Channel Sector Decomposition: Kitchen Sink (M3B) and Alternating (M3C) Specifications

	(1)	(2)
	M3B: Kitchen Sink	M3C: Alternating
<i>Panel A. Homogeneous structural predictors</i>		
HHI	−0.0996*** (−5.65)	−0.0174 (−1.41)
Supplier HHI	0.0281 (0.57)	0.0281 (0.50)
<i>Panel B. Sector × Supplier HHI interactions</i>		
SuppHHI × Real Estate	0.406*** (4.06)	0.362*** (3.26)
SuppHHI × Financials	0.298*** (3.02)	0.078 (0.89)
SuppHHI × Health Care	0.195*** (3.01)	0.062 (1.11)
SuppHHI × Industrials	0.118*** (3.33)	0.121** (2.38)
SuppHHI × Energy	0.055 (1.07)	0.084 (1.41)
SuppHHI × Materials	0.006 (0.13)	−0.018 (−0.41)
SuppHHI × Consumer Staples	−0.015 (−0.25)	−0.036 (−0.82)
SuppHHI × Technology	−0.016 (−0.27)	−0.072 (−1.40)
SuppHHI × Cons. Discr.	−0.030 (−0.51)	−0.085 (−1.46)
SuppHHI × Utilities	−0.070* (−1.68)	−0.088** (−2.14)
<i>Panel C. Controls</i>		
ATM IV	0.0016 (0.18)	0.0014 (0.18)
Total Return Index	1.60×10^{-5} (0.66)	2.24×10^{-5} (1.00)
Sector × HHI (M3B)	See App. Table 8	—
Entity FE	Yes	Yes
Time FE	Yes	Yes
Observations	2,425	2,425
R^2 (within)	0.005	−0.003
F -statistic	1.77	2.21
F p -value	0.012	0.006

Communication is the omitted reference. t -statistics in parentheses, entity-clustered.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Industrials carries exposure to single-source component suppliers, and Real Estate is sensitive to concentrated construction and financing inputs. All four are O-ring bottlenecks where the failure of a single upstream node can halt downstream operations. Model 3C confirms only Real Estate (+0.362, $t = 3.26$) and Industrials (+0.121, $t = 2.38$) under entity clustering, with the Financials and Health Care interactions losing significance under the alternating restriction.

Two further observations qualify this reading. The Supplier HHI interactions for Consumer Discretionary, Consumer Staples, Materials, Technology, and Utilities are not significant at the 5% level ($|t| < 1.7$ in all cases under M3B clustering), and several carry negative point estimates. For these sectors, upstream concentration does not generate predictable forward repricing of tail risk. Two readings are consistent with this: higher effective substitutability in their input baskets, or more efficient market pricing of their supply-chain exposures. Separately, the M3B HHI sector interactions, reported in Appendix Table 8 alongside their M3A counterparts, are stable in sign and magnitude relative to M3A for eight of the ten sectors. The Financials HHI interaction reverses sign between the two specifications, becoming significantly negative under entity clustering in M3B, and the Health Care HHI interaction attenuates from significant to insignificant. Both shifts are consistent with the multicollinearity that arises when heterogeneous internal HHI and heterogeneous Supplier HHI columns are included simultaneously, since the saturated M3B specification distributes a single underlying loading across two collinear regressors and produces unstable point estimates on each. The seven other HHI sector interactions retain their sign and approximate magnitude across the two specifications.

The bandwidth and stability diagnostics warrant explicit acknowledgment. All four M3B Supplier HHI sector interactions and both M3C Supplier HHI sector interactions lose significance under Driscoll-Kraay standard errors at a six-month bandwidth, and they exhibit fold-to-fold instability under the leave-one-year-out cross-validation reported in Appendix B, with same-sign frequency of 67% to 80% across folds and 5%-level significance in 13% to 27%

of folds. We therefore read M3B and M3C as exploratory rather than confirmatory specifications. They identify a candidate set of sectors where the upstream channel may operate independently of the internal channel, but the specific sector-by-sector point estimates should not be relied on for inference.

5.4 Out-of-Sample Forecasting

To guard against in-sample overfitting, we complement the inference models with an out-of-sample forecasting exercise. The training window spans January 2005 through December 2019, and the test window covers January 2025 through February 2026, with the intervening COVID-era period (2020–2024) excluded from both samples. Because the OptionMetrics download terminates on August 29, 2025 and the one-period forward shift truncates the final observation, the realized test sample contains seven months of forecasts (January through July 2025) per sector ETF, yielding 77 forecast observations across the eleven sectors. The baseline forecasting model includes only the macroeconomic controls (term spread, credit spread) and sector-level controls (ATM IV, total return index) with entity fixed effects. The challenger models augment this baseline with the structural concentration variables under three nested specifications. These are the Homogeneous (HHI plus Supplier HHI), the Heterogeneous HHI (Model 3A challenger), and the Kitchen Sink (Model 3B challenger). Statistical significance of the forecast-error differentials is assessed via the Diebold-Mariano test on monthly-averaged squared error differentials.

Table 5: Out-of-Sample Forecast Comparison: Mean Squared Prediction Error

Model	N forecasts	MSPE ($\times 10^4$)	% Δ vs. Baseline	DM statistic	DM p -value
Baseline	77	0.78	—	—	—
Homogeneous (HHI + Supp. HHI)	77	0.82	+5.2%	−2.52	0.012
Heterogeneous HHI (3A challenger)	77	0.99	+26.9%	−1.41	0.159
Kitchen Sink (3B challenger)	77	1.12	+43.8%	−1.58	0.114

Training: Jan 2005–Dec 2019; test: Jan–Jul 2025; 2020–2024 excluded. Entity FE only. Diebold–Mariano statistic on monthly-averaged squared error differentials, $h = 1$; negative DM indicates higher MSPE for the challenger.

Table 5 reports the comparison. None of the structural challenger models improves on the baseline, and the Homogeneous specification is in fact significantly worse at the 5% level (Diebold-Mariano $p = 0.012$). The Heterogeneous and Kitchen Sink challengers carry larger MSPE deteriorations in point estimate but do not reject equivalence formally. We do not read this as a failure of the structural mechanism. Instead, it is evidence that the in-sample inference is not the result of overfitting to a pattern that would also produce monthly out-of-sample forecasting gains. The gap between in-sample structural inference and monthly out-of-sample predictability is a well-documented feature of the equity premium prediction literature, and our results sit within that pattern. The structural concentration variables contain genuinely predictive information about the conditional shape of the forward risk reversal distribution, but at monthly frequency this information does not translate into point-forecast accuracy gains over a baseline that already includes ATM IV and macro controls. Section 5.5 returns to this distinction.

5.5 Discussion

5.5.1 Non-Monotonic Fragility and the Substitutability Gradient

The central empirical finding is that the relationship between concentration and forward tail-risk pricing is not monotone in either dimension. In the homogeneous specifications (Models 1 and the homogeneous control in M3A), the negative coefficient on Supplier HHI captures the variance-compression branch of the non-monotonic relationship. As upstream concentration rises within the regime where shocks remain in the ordinary range, ordinary-shock variance falls, and the options market correctly prices the lower variance with a less negative forward risk reversal. This is consistent with the diagonal-curvature result in equation (4) once one allows the second-order term to dominate first-order variance compression in the cross-section average. The market is not necessarily complacent. It is pricing what the realized shock distribution puts in front of it.

The sector-heterogeneous specifications surface the tail-risk branch of the same mechanism. The seven sectors with positive net loadings on internal HHI under Model 3A (Real Estate, Health Care, Utilities, Materials, Industrials, Energy, Consumer Staples) are those where physical production processes involve long lead times, specialized capital goods, and single-source dependencies that render inputs effectively non-substitutable in the short run. In the language of the theory, these are sectors where the endogenous formation of supplier networks under fixed costs of qualification has driven $\sigma_{\text{eff}} \rightarrow 0$, maximizing the non-linear amplification term $\lambda_i(1 - \lambda_i)(\sigma - 1)$ in equation (4). By contrast, Technology and Communication operate in production environments closer to the unit-elasticity case where the amplification term collapses, and their net loadings are correspondingly negative or insignificant. The substitutability gradient in Figure 3 is the direct empirical fingerprint of this theoretical prediction. The options market, at least implicitly, distinguishes between the two regimes.

The upstream-channel diagnostics in Models 3B and 3C tell a complementary story under the same logic. The four sectors where Supplier HHI sector interactions are largest under M3B clustering (Financials, Health Care, Industrials, Real Estate) are precisely those whose upstream supply chains exhibit O-ring bottleneck configurations, although the bandwidth sensitivity and LOO instability of these point estimates argue for caution in pressing the sector-by-sector reading too hard. The insignificance of the upstream channel for Technology, Consumer Discretionary, and Consumer Staples is consistent with these industries sourcing from sufficiently diversified upstream baskets to avoid single-point-of-failure vulnerabilities, or with the market pricing those exposures more efficiently in higher-profile sectors.

5.5.2 The Hub Shock Channel as Secondary

The Hub Shock specification in Model 2 receives partial empirical support. The additive Hub Shock coefficient is significant at the 5% level under entity clustering ($t = 2.53$) and marginal under Driscoll-Kraay ($t = 1.90$), suggesting that acute distress in heavyweight con-

stituents prompts a modest level shift in next-month forward skew. The earlier interactive formulation testing whether internal concentration amplifies the Hub Shock effect ($t = -0.41$ on $HHI \times \text{HubShock}$) provided no support for joint amplification. Hub Shock and structural concentration appear to operate as separate transmission channels rather than as a coupled mechanism, and the Hub Shock channel is the smaller of the two. This is consistent with a reading in which the slow-moving structural channel reflects the network curvature mechanism in equation (4) while Hub Shock proxies for a residual high-frequency volatility-spillover component. The non-monotonic fragility framing does not require the cascade-style joint amplification that an earlier draft of this paper hypothesized.

5.5.3 The In-Sample, Out-of-Sample Distinction

The OOS exercise is best read as a defensive specification rather than as a forecasting claim. The structural challengers do not improve point-forecast accuracy at monthly horizons, and the Homogeneous challenger underperforms the baseline statistically. We do not read this as a rejection of the structural mechanism. Two observations support the inference reading. First, if the in-sample coefficients were the artifact of overfitting to a feature in the training data, the structural challenger should have exploited that feature in the test window and produced an MSPE *improvement* over a baseline that does not see the structural variables. The opposite pattern in Table 5 indicates that the in-sample relationships are not driven by a forecastable feature of the conditional mean. Second, the broader equity-premium prediction literature documents an analogous gap between in-sample structural inference and out-of-sample point-forecast gains. Concentration variables that carry meaningful information about the conditional distribution of forward tail risk need not improve squared-error prediction over a baseline that already absorbs the level of the volatility regime through ATM IV. Future work targeting realized-skewness validation or distributional forecasts would be a more direct test of the structural channel than monthly point-forecast MSPE.

5.5.4 Limitations and Directions for Future Research

Several features of the empirical design impose boundaries on the interpretation of these results. First, the BEA-to-GICS concordance underlying the Supplier HHI construction relies on manual keyword matching, introducing potential measurement error into the upstream concentration proxy. This attenuation bias works against finding significant results and therefore does not inflate our estimates, but a more granular concordance using firm-level supply chain disclosures could sharpen the channel identification. Recent work using detailed customer-supplier disclosures (Carvalho and Tahbaz-Salehi, 2019) suggests that node-level resolution would reveal additional structure that sector-level aggregation washes out.

Second, the substitutability gradient is identified through the theoretical correlation between sector identity and the effective elasticity of substitution rather than through a direct empirical measure of σ_{eff} . Direct empirical estimation of sector-level effective elasticities of substitution, combined with Domar-weight integration from BEA gross output data, would allow a direct test of equation (4) that does not rely on sector dummies as proxies. We see this as the most promising near-term extension of the present design.

Third, the bandwidth-sensitivity of the M3B and M3C upstream-channel sector interactions, combined with the Real Estate within-panel-length issue, places real boundaries on the precision of the sector-by-sector inference. The robustness appendix documents these limitations transparently, and the headline finding (the substitutability gradient under Model 3A across the seven physical-capital-intensive sectors and the digital sectors) survives both DK6 and LOO. The within-sector point estimates that do not survive these checks should be read as suggestive rather than confirmatory.

Fourth, we drop the 2020–2024 period from the out-of-sample evaluation window because of the extraordinary nature of the COVID-19 regime and the post-pandemic monetary-policy environment. This leaves open the question of whether the concentration-to-skew relationship holds during periods of acute systemic stress. The COVID-19 pandemic simultaneously disrupted physical supply chains and repriced tail risk across all sectors, creating a natural

experiment that could provide powerful identification. A clean test of the non-monotonic mechanism through that window would require careful treatment of the confounding aggregate shock, and we leave it for future work.

Finally, the present analysis operates at the sector level, which limits the granularity with which production network topology can be observed. A firm-level extension using customer-supplier relationship disclosures and plant-level geographic data could test whether the substitutability-gradient mechanism operates with greater precision when measured at the node level rather than the sector level. Such an extension would also permit a more direct empirical test of how thresholds in the Acemoglu et al. (2015) phase-transition theorem map to observable network topology, which is one piece of the non-monotonic story that the present design can diagnose only through the sectoral substitutability gradient rather than through a direct empirical location of the threshold.

6 Conclusion

This paper has tested whether rising concentration in critical, high-Domar-weight sectors leads the repricing of forward tail risk in equity options markets. The empirical evidence supports an affirmative answer, qualified by a substantive economic structure. The relationship between concentration and the forward 25-delta risk reversal is not monotone. Its sign and magnitude depend on whether the relevant concentration is measured within a sector or inherited from upstream, and on the effective elasticity of substitution governing the sector’s input bundle.

Three findings anchor the paper’s argument. First, the Input-Output-weighted Supplier HHI is the dominant homogeneous predictor of forward tail-risk pricing in the baseline structural model, with a coefficient of -0.076 ($t = -5.22$). Read through the lens of the non-monotonic fragility mechanism formalized in Section 3 and equation (6), this negative loading is consistent with the variance-compression branch of the relationship. Hub formation re-

duces vulnerability to ordinary shocks while concentrating latent exposure in the extreme tail. Second, the sector-heterogeneous specifications recover the substitutability gradient predicted by the diagonal-curvature result of equation (4). Seven physical-capital-intensive sectors (Real Estate, Health Care, Utilities, Materials, Industrials, Energy, and Consumer Staples) load positively on internal HHI under Model 3A, while the digital and platform-economy sectors (Technology and Communication) load negatively. Economic magnitudes range from roughly +10 basis points per one-standard-deviation increase in HHI in Real Estate and Energy down to negative or near-zero values in the platform sectors. Third, the upstream channel is itself sector-heterogeneous. The Model 3B and Model 3C diagnostics flag Real Estate and Industrials, and more tentatively Financials and Health Care, as candidate sectors where inherited supplier concentration carries independent forward-looking content beyond the within-sector channel.

A secondary channel in Model 2 complements these structural findings without competing with them. Volatility spikes in heavyweight constituents that are currently central to their sector produce a modest additive upward adjustment in next-month forward skew, with a Hub Shock coefficient of 0.0014 that is significant at the 5% level under entity clustering ($t = 2.53$) and marginal under Driscoll-Kraay at six months ($t = 1.90$). An earlier interactive formulation between HHI and Hub Shock provided no support for joint amplification ($t = -0.41$). We therefore read Hub Shock as an independent high-frequency level shift rather than as an amplifier of the slow-moving structural mechanism.

Taken together, these findings give empirical content to the prediction of Baqaee and Farhi (2019) that complementarities, not concentration alone, generate non-linear amplification of downside risk. They extend the network-pricing logic of Herskovic (2018) from average expected returns to the asymmetric tail-pricing moment of the equity option surface.

The results carry two practical implications for risk pricing and macroprudential monitoring. First, market participants who track at-the-money implied volatility as a summary statistic for the forward-looking risk environment are absorbing the level of the volatility

regime but not the structural composition of the production network that conditions its tails. The 25-delta risk reversal contains information about latent network fragility that ATM IV does not, and that information is partially predictable from observable concentration measures. Second, the substitutability gradient suggests that risk managers and supervisors should treat concentration in physical-capital-intensive sectors and concentration in digital sectors as economically distinct objects. A given level of HHI in a Materials or Energy sector implies more latent left-tail exposure than the same HHI in a Technology or Communication sector, and the difference is large enough to matter for hedging and capital-allocation decisions. Monitoring upstream Supplier HHI provides a complementary, slow-moving leading indicator that aggregates fragility information across the inter-sector dependency structure rather than within any single industry.

Several limitations bound the strength of the inference. The bandwidth-sensitivity of the Model 3B and Model 3C upstream-channel sector interactions, the relatively short within-sector panel for Real Estate, and the absence of the 2020–2024 period from the out-of-sample evaluation window each constrain how aggressively the sector-by-sector point estimates can be read. The headline finding (the substitutability gradient under Model 3A across the seven physical-capital-intensive sectors and the two clearly digital sectors) survives both Driscoll-Kraay inference at six- and twelve-month bandwidths and the leave-one-year-out cross-validation reported in Appendix B. Health Care, Industrials, and Energy are robust under both inference regimes, and Materials and Consumer Staples are robust on the dimensions separately tested. The granular within-sector decomposition of the upstream channel should nevertheless be read as exploratory rather than confirmatory, and the Real Estate point estimate is best read as directionally consistent with the gradient rather than as a precisely identified magnitude. The out-of-sample exercise documents that the structural variables do not improve monthly point-forecast accuracy over a baseline that already includes ATM IV and macroeconomic controls. This pattern is consistent with the broader equity-premium prediction literature and not with overfitting, and it reinforces the primary

use of the structural variables as inferential rather than predictive instruments at the monthly horizon.

Several near-term extensions follow naturally from the present design. Direct empirical estimation of sector-level effective elasticities of substitution, combined with Domar-weight integration from BEA gross-output data, would replace the sector-dummy proxy used here with a quantitative measure of σ_{eff} and provide a direct test of equation (4). Realized-skewness validation against the implied-volatility-based dependent variable would offer a second window on the structural channel that does not rely on monthly point-forecast accuracy. A firm-level extension using customer-supplier disclosure data would resolve network topology at a finer grain than sector-level concentration permits, allowing a more direct empirical localization of the threshold $\bar{\epsilon}$ in the phase-transition relation of equation (6). Methods for empirically locating non-linear thresholds, drawing on regime-switching and threshold-regression diagnostics, would tighten the identification of the boundary between the stabilizing and amplifying branches of the non-monotonic mechanism. Each of these extensions targets a specific piece of the structural framework that the current sector-level design can probe only indirectly through the substitutability gradient.

The motivating intuition can be put simply. When supply chains thin out and concentrate around a small number of essential nodes, the production network becomes more efficient at absorbing routine fluctuations and more vulnerable to the kind of acute hub failure that shows up in the equity skew rather than in the equity variance. That asymmetric pricing pattern is the empirical fingerprint of a particular kind of structural risk, and it appears most clearly in the sectors where the underlying physical economics most strongly predicts it.

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A Second-Order Approximation of Aggregate Output and Endogenous Skewness

Following the framework of Baqaee and Farhi (2019), this appendix derives the second-order response of aggregate output to microeconomic shocks under a Constant Elasticity of Substitution (CES) production structure. We show that when inputs are strict complements ($\sigma < 1$), the economy structurally generates negative skewness. For tractability, we work in a simplified single-tier CES economy where labor is the only primary factor and is supplied inelastically. The result extends to the full multi-sector input-output network of Baqaee and Farhi (2019), but that generalization is not reproduced here.

A.1 The Baseline: Hulten’s Theorem

Let $A_i > 0$ denote the total factor productivity of sector i , and define $\epsilon_i \equiv \ln A_i$ as its log-productivity level. Let $y \equiv \ln Y$ denote log aggregate real output. When evaluating deviations around a symmetric steady state in which $\mathbb{E}[\ln A_i] = 0$ for all i , ϵ_i coincides with the mean-zero deviation $\epsilon_i = \ln A_i - \mathbb{E}[\ln A_i]$ used in the main text.¹²

Under the assumptions of competitive equilibrium, constant returns to scale, and a first-order approximation, Hulten’s Theorem (Hulten, 1978) states that the elasticity of aggregate output with respect to a sectoral productivity shock equals that sector’s Domar weight:

$$\frac{\partial y}{\partial \epsilon_i} = \lambda_i, \quad \lambda_i \equiv \frac{P_i Y_i}{PY}, \quad (17)$$

where λ_i is the ratio of sector i ’s nominal gross sales to nominal GDP. Under this linear approximation, positive and negative shocks of equal magnitude have symmetric effects on aggregate output.

¹²The level and deviation definitions are numerically identical under the normalization $\mathbb{E}[\ln A_i] = 0$, which is standard in this literature (Baqaee and Farhi, 2019).

A.2 The Second-Order Expansion

Because equation (17) is a first-order approximation, it fails to capture the non-linear amplification of shocks in networks with rigidities. To capture structural skewness, we take a second-order Taylor expansion of $\Delta y \equiv y - \mathbb{E}[y]$ around the steady state $\epsilon_i = 0$ for all i :

$$\Delta y \approx \sum_i \lambda_i \epsilon_i + \frac{1}{2} \sum_{i,j} \frac{\partial^2 y}{\partial \epsilon_i \partial \epsilon_j} \epsilon_i \epsilon_j. \quad (18)$$

Focusing on the idiosyncratic volatility of a single hub sector i , we require the diagonal second derivative $\frac{\partial^2 y}{\partial \epsilon_i^2}$. Differentiating both sides of Hulten's result (17) with respect to ϵ_i a second time, and recognizing that the Domar weight λ_i is itself a function of the shock vector $(\epsilon_1, \dots, \epsilon_N)$, yields:

$$\frac{\partial^2 y}{\partial \epsilon_i^2} = \frac{\partial}{\partial \epsilon_i} \left(\frac{\partial y}{\partial \epsilon_i} \right) = \frac{\partial \lambda_i}{\partial \epsilon_i}. \quad (19)$$

The problem therefore reduces to computing how a sector's Domar weight responds to its own productivity shock.

A.3 Domar Weights in a CES Economy

We work in a simplified single-tier CES economy with labor as the only primary factor, supplied inelastically, and normalize the wage to unity ($W = 1$). We further assume each sector employs a unit labor requirement per unit of output, so that the marginal cost, and hence the competitive price, of good i is:

$$P_i = \frac{W}{A_i} = e^{-\epsilon_i}. \quad (20)$$

Sector weights $\omega_i > 0$ in the CES aggregator correspond to the quality/efficiency parameters α_{ij} of the main text (equation (1)) in the special case where each downstream firm sources

from exactly one upstream sector. The aggregate ideal price index is:

$$P = \left(\sum_j \omega_j P_j^{1-\sigma} \right)^{1/(1-\sigma)}. \quad (21)$$

Using standard CES demand, the Domar weight for sector i is the expenditure share:

$$\lambda_i = \frac{\omega_i P_i^{1-\sigma}}{\sum_j \omega_j P_j^{1-\sigma}} = \omega_i \left(\frac{P_i}{P} \right)^{1-\sigma}. \quad (22)$$

A.4 Deriving the Network Curvature

Substituting equation (20) into (22) and taking the natural logarithm:

$$\ln \lambda_i = \ln \omega_i - (1 - \sigma)\epsilon_i - (1 - \sigma) \ln P. \quad (23)$$

Differentiating with respect to ϵ_i :

$$\frac{\partial \ln \lambda_i}{\partial \epsilon_i} = -(1 - \sigma) - (1 - \sigma) \frac{\partial \ln P}{\partial \epsilon_i}. \quad (24)$$

We evaluate $\frac{\partial \ln P}{\partial \epsilon_i}$ from the price index (21). Since $\frac{\partial P_i^{1-\sigma}}{\partial \epsilon_i} = -(1 - \sigma) P_i^{1-\sigma}$ from equation (20):

$$\frac{\partial \ln P}{\partial \epsilon_i} = \frac{-\omega_i P_i^{1-\sigma}}{\sum_j \omega_j P_j^{1-\sigma}} = -\lambda_i. \quad (25)$$

Substituting (25) into (24) and simplifying:

$$\frac{\partial \ln \lambda_i}{\partial \epsilon_i} = -(1 - \sigma) + \lambda_i(1 - \sigma) = (\sigma - 1)(1 - \lambda_i). \quad (26)$$

Applying $\frac{\partial \lambda_i}{\partial \epsilon_i} = \lambda_i \cdot \frac{\partial \ln \lambda_i}{\partial \epsilon_i}$ and substituting into equation (19):

$$\frac{\partial^2 y}{\partial \epsilon_i^2} = \lambda_i (1 - \lambda_i) (\sigma - 1). \quad (27)$$

A.5 Conclusion

Equation (27) proves the mechanical generation of aggregate skewness. When inputs are strict complements ($\sigma < 1$), the term $(\sigma - 1)$ is strictly negative. Consequently, a negative productivity shock to a hub sector causes its Domar weight λ_i to *increase* endogenously: as the hub’s price spikes and substitution fails, downstream expenditure is forced onto the now-scarcer input, amplifying rather than absorbing the shock. This yields a strictly negative second derivative in equation (18) and translates symmetric microeconomic shocks into asymmetric, negatively skewed macroeconomic outcomes.

The magnitude of this effect scales directly with the Herfindahl–Hirschman Index. Since $HHI \equiv \sum_i \lambda_i^2$, a more concentrated network places greater weight on sectors with large λ_i , for which the term $\lambda_i(1 - \lambda_i)$ in equation (27) is maximized. This closes the theoretical chain from the empirical measure HHI to the physical generation of left-tail risk in aggregate output.

B Robustness Appendix

This appendix documents the inference robustness exercises that underpin the headline findings reported in Section 5. Two exercises are reported. The first applies Driscoll–Kraay standard errors at six- and twelve-month bandwidths to all coefficients carrying interpretive weight in the main text, accommodating both within-entity serial correlation and contemporaneous cross-sectional dependence. The second runs a leave-one-year-out cross-validation across the fifteen calendar years of the in-sample training window, reporting fold-to-fold stability of sign and statistical significance for the same set of headline coefficients. A short closing subsection explains the design issues that led us to drop the originally planned expanding-window out-of-sample exercise in favor of the fixed-window design reported in Section 5.4.

B.1 Driscoll–Kraay Standard Errors

The headline specifications cluster standard errors at the entity level to absorb within-sector serial correlation. Driscoll and Kraay (1998) standard errors generalize this restriction to allow for both serial correlation within entities and contemporaneous cross-sectional dependence among entities. The latter is plausible in production-network settings where common upstream shocks generate correlated residuals across downstream sectors. We report DK standard errors at bandwidths of six and twelve months, following the convention of choosing the bandwidth roughly equal to the square root of the time dimension and a longer alternative as a sensitivity check on the kernel decay rate.

Table 6 reports clustered, DK6, and DK12 t -statistics for every coefficient that carries headline interpretive weight in Section 5. Three patterns are worth emphasizing. First, the M1 Supplier HHI coefficient remains significant at the 5% level under both DK bandwidths (DK6 $t = -2.43$, DK12 $t = -2.44$), confirming that the variance-compression branch of the non-monotonic mechanism is not an artifact of the within-entity clustering structure. Second, the M3A internal-HHI substitutability gradient is bandwidth-robust for the core of the gradient. Energy, Health Care, Industrials, and Materials retain 5% significance under DK6, the Consumer Staples interaction retains 10% significance, the M3A baseline HHI loading retains 1% significance, and the homogeneous Supplier HHI in M3A retains 10% significance under both DK bandwidths. Third, the M3B Kitchen Sink and M3C Alternating upstream-channel sector interactions, which are highly significant under entity clustering at the 1% level for five of six rows reported, lose 5% significance under DK6 across the board. The Real Estate HHI interaction in M3A and the Consumer Discretionary HHI interaction in M3A also lose 5% significance under DK6. The former is due to the relatively short within-sector panel for XLRE, the latter because the entity-clustered standard error appears to understate the cross-sectional dependence in the residual for that interaction.

The Hub Shock additive coefficient in Model 2 displays the bandwidth sensitivity flagged

Table 6: Driscoll–Kraay Robustness: Clustered, DK6, and DK12 t -Statistics

	Coef.	t (Clustered)	t (DK6)	t (DK12)
<i>Panel A. Homogeneous structural variables</i>				
M1: HHI	−0.0073	−1.06	−0.70	−0.74
M1: Supplier HHI	−0.0758	−5.22***	−2.43**	−2.44**
M2: HHI	0.0023	0.21	0.22	0.21
M2: Hub Shock	0.0014	2.53**	1.90*	1.82*
M3A: HHI	−0.0859	−4.94***	−2.59***	−2.67***
M3A: Supplier HHI	−0.0632	−4.65***	−1.78*	−1.80*
M3B: HHI	−0.0996	−5.65***	−3.07***	−3.16***
M3B: Supplier HHI	0.0281	0.57	0.28	0.32
M3C: HHI	−0.0174	−1.41	−1.18	−1.24
M3C: Supplier HHI	0.0281	0.50	0.32	0.36
<i>Panel B. M3A heterogeneous HHI sector interactions (ref. Communication)</i>				
HHI $\times$ Real Estate	0.298	3.52***	0.95	0.90
HHI $\times$ Health Care	0.163	3.18***	2.29**	2.24**
HHI $\times$ Utilities	0.138	1.35	0.67	0.60
HHI $\times$ Materials	0.125	4.29***	2.18**	1.93*
HHI $\times$ Industrials	0.110	5.68***	2.54**	2.64***
HHI $\times$ Energy	0.107	4.49***	2.94***	2.98***
HHI $\times$ Consumer Staples	0.101	3.90***	1.90*	1.81*
HHI $\times$ Consumer Discretionary	0.059	3.20***	1.60	1.65*
HHI $\times$ Financials	0.039	0.51	0.23	0.20
HHI $\times$ Technology	0.013	0.51	0.24	0.23
<i>Panel C. M3B and M3C upstream-channel sector interactions</i>				
M3B: SuppHHI $\times$ Real Estate	0.406	4.06***	0.77	0.93
M3B: SuppHHI $\times$ Financials	0.298	3.02***	1.59	1.76*
M3B: SuppHHI $\times$ Health Care	0.195	3.01***	1.29	1.23
M3B: SuppHHI $\times$ Industrials	0.118	3.33***	0.98	0.96
M3C: SuppHHI $\times$ Real Estate	0.362	3.26***	0.70	0.88
M3C: SuppHHI $\times$ Industrials	0.121	2.38**	1.15	1.13

Clustered t -statistics use entity-clustered standard errors. DK6 and DK12 use Driscoll–Kraay standard errors with Bartlett kernel and bandwidths of six and twelve months. Significance markers attach to the t -statistics. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

in Section 5.2. The clustered t -statistic of 2.53 attenuates to 1.90 under DK6 and to 1.82 under DK12, retaining 10% significance under both DK bandwidths but losing 5% significance. This pattern is consistent with the Hub Shock channel operating as a secondary additive level shift rather than a structural mechanism. It supports the choice in the main text to treat Hub Shock as a complementary but non-central transmission channel relative to the slow-moving structural concentration variables.

B.2 Leave-One-Year-Out Cross-Validation

The LOO exercise re-estimates each headline specification fifteen times on the in-sample training window (January 2005 through December 2019), dropping one calendar year from the estimation sample on each fold and recomputing the coefficient and its t -statistic on the remaining fourteen-year subsample. The exercise targets two stability properties. The first is sign stability, measured as the share of folds in which the re-estimated coefficient retains the same sign as the full-sample estimate. The second is fold-level significance, measured as the share of folds in which the re-estimated coefficient remains significant at the 5% level under entity-clustered standard errors. We restrict attention to the variables that the Driscoll–Kraay exercise flags for stability scrutiny.

Table 7 reports the results. The M1 Supplier HHI coefficient is uniformly stable, retaining the negative sign in 100% of folds and remaining significant at the 5% level in 100% of folds, with an interquartile range across folds that is small relative to the full-sample point estimate. The M3A internal-HHI baseline coefficient is also highly stable, retaining its negative sign in 93% of folds and 5% significance in 80% of folds. Among the M3A sector interactions, Health Care, Industrials, and Energy each retain their positive sign in 93% of folds and 5% significance in 80% of folds. These three diagnostics align with the substitutability gradient surviving fold-to-fold variation in the training data.

The instability identified by the Driscoll–Kraay exercise is also visible in the LOO. The Real Estate HHI interaction in M3A retains its sign in 93% of folds but is significant at the

Table 7: Leave-One-Year-Out Cross-Validation Stability (15 Folds, 2005–2019)

Variable	Full-sample coef.	Median coef.	Same-sign share (%)	5% sig. share (%)
<i>Panel A. Headline homogeneous structural variables</i>				
M1: Supplier HHI	−0.069	−0.071	100.0	100.0
M3A: HHI (baseline)	−0.286	−0.284	93.3	80.0
M3A: Supplier HHI	−0.066	−0.066	100.0	93.3
<i>Panel B. M3A heterogeneous HHI sector interactions</i>				
HHI $\times$ Health Care	0.372	0.373	93.3	80.0
HHI $\times$ Industrials	0.316	0.320	93.3	80.0
HHI $\times$ Energy	0.305	0.302	93.3	80.0
HHI $\times$ Real Estate	0.705	0.686	93.3	26.7
<i>Panel C. M3B and M3C upstream-channel sector interactions</i>				
M3B: SuppHHI $\times$ Real Estate	0.340	0.339	80.0	26.7
M3B: SuppHHI $\times$ Financials	0.578	0.577	93.3	13.3
M3B: SuppHHI $\times$ Health Care	0.491	0.500	93.3	13.3
M3B: SuppHHI $\times$ Industrials	0.329	0.332	93.3	13.3
M3C: SuppHHI $\times$ Real Estate	0.728	0.725	80.0	93.3
M3C: SuppHHI $\times$ Industrials	0.605	0.616	93.3	66.7

Re-estimation on the 2005–2019 training window with one calendar year held out per fold (15 folds), entity-clustered standard errors. Same-sign share is the percentage of folds in which the re-estimated coefficient retains the sign of the full-sample LOO estimate; 5% sig. share is the percentage significant at the 5% level. Coefficients are estimated on 2005–2019 and may differ in magnitude from full-sample (2005–February 2026) figures in Tables 1 and 4.

5% level in only 27% of folds, with a coefficient range that includes large negative excursions in folds where the limited XLRE within-sector panel is further reduced by holding out one of its training years. The four M3B upstream-channel sector interactions retain the same sign in 80% to 93% of folds but reach 5% significance in only 13% to 27% of folds, consistent with the bandwidth attenuation documented in Table 6.

The two M3C upstream-channel sector interactions are a partial exception. They retain the same sign in 80% and 93% of folds and reach 5% significance in 93% and 67% of folds, respectively. The alternating restriction in M3C, which holds internal HHI homogeneous while interacting Supplier HHI with sector dummies, recovers more stable upstream-channel point estimates under LOO than the fully saturated kitchen-sink specification in M3B. Both specifications nonetheless lose significance under Driscoll–Kraay inference, so the M3C LOO stability is best read as a diagnostic that the Real Estate and Industrials upstream loadings are not artifacts of any single year in the training data, rather than as a contradiction of the bandwidth-attenuation finding.

B.3 Joint Reading

The two robustness exercises track partially overlapping sets of variables, so the joint reading must distinguish between coefficients tested under both regimes and coefficients tested under one. The M1 Supplier HHI coefficient, the M3A baseline HHI coefficient, the M3A homogeneous Supplier HHI coefficient, and the Health Care, Industrials, and Energy sector HHI interactions in M3A are robust under both DK6 (at the 5% level, except for M3A Supplier HHI which is significant at the 10% level under DK6) and LOO (5% significance in 80% or more of folds). The M3B baseline HHI coefficient and the Materials sector HHI interaction in M3A retain 5% DK6 significance and were not separately tracked in the LOO exercise, so they are robust on the dimension tested. The Consumer Staples HHI interaction in M3A retains 10% DK6 significance and was likewise not tracked in the LOO. Findings losing 5% significance under DK6 include the Real Estate HHI interaction in M3A, the Consumer Dis-

cretionary HHI interaction in M3A, all four M3B Supplier HHI sector interactions, and both M3C Supplier HHI sector interactions.

The headline substitutability-gradient finding under Model 3A is supported by the three sector interactions tracked under both robustness regimes (Health Care, Industrials, Energy), reinforced by the M3A baseline HHI loading itself, and corroborated by the DK6-only check on Materials and Consumer Staples. The Real Estate point estimate fails both robustness checks at the 5% level and is best read as directionally consistent with the gradient rather than as a precisely identified magnitude. The granular within-sector decomposition of the upstream channel under Models 3B and 3C does not survive DK6 at the 5% level for any of the six sector interactions reported, and is treated in the main text as exploratory rather than confirmatory. The M3B and M3C estimates serve to identify candidate sectors where inherited supplier concentration plausibly carries forward-looking content beyond the within-sector channel.

B.4 Note on the Dropped Expanding-Window Out-of-Sample Exercise

The fixed-window OOS design reported in Section 5.4 was preceded by an expanding-window design with an initial training cutoff of December 2014. The training sample was incremented one month at a time and the next-month forecast was generated from the re-estimated model. The post-sample launches of XLRE in late 2015 and XLC in 2018 generated 43 to 44 absorbing-effect failures across the early expansion folds. The entity fixed effect for the newly entering sector ETF could not be identified from a single training observation, and the next-period forecast collapsed mechanically. We therefore abandoned the expanding-window specification. The fixed-window design reported in Section 5.4 sidesteps this issue by training on a single contiguous block of pre-pandemic data in which all eleven entity fixed effects are identified throughout, and by evaluating on a single contiguous block of post-pandemic data in which all entity fixed effects can be projected forward without re-

estimation. The trade-off is a smaller, fixed test window of seven monthly forecasts per ETF rather than a steadily growing one, and the resulting Diebold–Mariano test in Table 5 is correspondingly less powerful than the expanding-window design would have been in the absence of the absorbing-effect failures.

C Extended Results

This appendix houses the Model 3B internal-HHI sector interactions referenced in the footer of main Table 4. The substantive purpose of the table is to display the full set of $\text{HHI} \times \text{sector}$ loadings under the fully saturated specification of equation (14), alongside the corresponding M3A loadings, so the reader can directly compare the heterogeneous internal-HHI structure across the two specifications. The discussion of multicollinearity-induced sign and magnitude shifts in Section 5.3.2 of the main text refers to this table.

The seven sectors with stable HHI sector interactions across M3A and M3B (Real Estate, Materials, Industrials, Energy, Consumer Staples, Consumer Discretionary, and Technology) are those for which the substitutability-gradient reading in Section 5.3.1 holds independently of whether the Supplier HHI channel is constrained to a homogeneous coefficient or allowed to vary by sector. The Health Care and Financials interactions, by contrast, are sensitive to the saturation choice. The M3B coefficients are absorbed in part by the sector-specific Supplier HHI interactions reported in main Table 4. The Utilities HHI interaction tightens from $+0.138$ ($t = 1.35$) under M3A to $+0.280$ ($t = 2.65$) under M3B, lifting it into the cluster of significantly positive sectors under the saturated specification. The M3A reading remains the headline because the Driscoll–Kraay and leave-one-year-out diagnostics in Appendix B apply to the M3A baseline rather than to the M3B sector-by-sector decomposition.

Table 8: M3B Internal-HHI Sector Interactions and Comparison with M3A

	(1)	(2)
	M3A interaction	M3B interaction
HHI $\times$ Real Estate	0.298*** (3.52)	0.333*** (3.56)
HHI $\times$ Health Care	0.163*** (3.18)	0.034 (0.44)
HHI $\times$ Utilities	0.138 (1.35)	0.280*** (2.65)
HHI $\times$ Materials	0.125*** (4.29)	0.132*** (6.21)
HHI $\times$ Industrials	0.110*** (5.68)	0.123*** (4.20)
HHI $\times$ Energy	0.107*** (4.49)	0.121*** (4.34)
HHI $\times$ Consumer Staples	0.101*** (3.90)	0.114*** (3.29)
HHI $\times$ Consumer Discretionary	0.059*** (3.20)	0.077*** (3.99)
HHI $\times$ Financials	0.039 (0.51)	-0.365*** (-2.86)
HHI $\times$ Technology	0.013 (0.51)	0.044 (1.53)
Homogeneous baseline HHI	-0.0859*** (-4.94)	-0.0996*** (-5.65)
Supplier HHI structure	Homogeneous	Het. by sector

Communication is the omitted reference. Each row reports the differential HHI loading for the named sector relative to Communication. t -statistics in parentheses, entity-clustered. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Code and Data Availability

All code, data construction scripts, and replication materials for this paper are available at <https://github.com/EthanSterling04/market-concentration-tail-risk>. The repository contains the data pipeline, regression code, and figure-generation scripts needed to reproduce the tables and figures reported in the main text and appendices.